

Interpreting Estimated Pricing Errors

Evidence from Characteristic-Based Return Forecasts

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27 September 2026

Abstract

What does a smaller estimated pricing error tell a model user? Using 153 U.S. stock characteristics, I separate changes in fitted expected returns from changes in factor loading spaces. A large difference in models' fitted characteristic-intercept magnitudes mainly reflects different estimated means. Historical residual diagnostics add little to forecast selection. Calibrating the residual component improves on the complete forecast in 20 of 24 specifications, but beats forecast averaging in eight and characteristic-principal-component calibration in six. These exploratory results distinguish explaining estimated pricing errors from using their characteristic component to predict subsequent clipped returns.

Keywords: pricing errors; characteristic alpha; factor models; forecast evaluation; model selection.

JEL classification: C38, C52, C53, G12.

1 Introduction

A researcher comparing factor models faces a basic interpretation problem. One model reports smaller estimated pricing errors: less of the estimated expected return remains outside its factor contribution. Does that model explain expected returns better, and should its forecasts replace those of the other model? These questions become difficult when each model estimates both the expected-return signal and the factor structure used to explain it. The reported discrepancy can fall because the explanatory structure changes, because the estimated signal changes, or both. This paper separates these changes and examines whether the resulting diagnostics help a user choose or adjust a return forecast.

A pricing error, or alpha, is an expected-excess-return discrepancy relative to a specified factor-pricing benchmark, information set and return horizon. It measures what that benchmark leaves unexplained. An estimate of the discrepancy also depends on how expected returns and the benchmark are estimated. A simple illustration shows the distinction. Suppose two fitted signals for one stock are 1.0% and 0.7% per month, while both assign 0.6% to the factor component. Their estimated residual components are 0.4% and 0.1%. The second is smaller because the complete signal is lower; this comparison alone says nothing about which signal is closer to the unknown expected return. These illustrative numbers are not empirical estimates. The application asks the analogous question across many stocks and an estimated factor loading space.

The empirical setting is a characteristic-based latent-factor model. Stock characteristics generate both factor exposures and an intercept component. Once the location of the latent factors is fixed, that intercept has a precise interpretation as the model's fitted characteristic alpha. The analysis concerns this characteristic-generated component, rather than every possible stock-level pricing error. Moreover, asking a model's loading space to explain a separate reference mean produces a projection residual, which need not be that model's own intercept. Keeping these objects distinct allows a controlled comparison without changing the meaning of alpha.

The application estimates models on the same observations of U.S. stocks with 153 characteristics. One estimator gives each stock the same within-month weight; the other gives larger stocks more weight in proportion to their formation-date market capitalization. These are called equal-weighted (EW) and value-weighted (VW) training. They are two ways to fit a forecasting model, not portfolio positions. Each produces a complete fitted mean and a loading space: the set of return signals that its estimated exposures can generate as factor contributions vary. Measurement weights are then held common within a comparison, so that a change in the assessment standard does not obscure the difference between the fitted

objects.

The first experiment crosses the two fitted means with the two loading spaces. Holding the mean fixed asks which space represents the same signal more closely. Holding the space fixed asks how much the residual changes when a different signal is supplied. Under equal-stock measurement, each model’s residual against its own mean has RMS 117.37 and 77.27 monthly basis points, respectively. Yet replacing only the space used to explain the EW mean changes its residual to 116.67 bp. Replacing only the mean within the EW space changes the residual to 75.02 bp. Thus the large difference between the two models’ reported residuals mainly accompanies a change in the estimated mean, rather than a substantial improvement in spanning a fixed mean. The result identifies the source of the reported contrast; subsequent returns are needed to assess the signals’ predictive accuracy.

The second experiment tests a possible use of the diagnostic: selecting one complete forecast before observing its outcomes. A smaller residual might indicate a useful factor representation, but it need not indicate a more accurate complete forecast. At nine annual origins, a comparison using each model’s own mean always selects the VW-trained model and therefore adds no selection information beyond that constant choice. Using the same historical ridge mean for both spaces changes the choices. Its EW prediction R^2 , however, is 0.3612%, compared with 0.4396% for an equal average of the two complete forecasts and 0.3972% for pooled ridge. The VW evaluation also favors those benchmarks. Fixing the target makes the diagnostic comparison clearer; its forecasting value remains an empirical question.

The third experiment concerns a different decision. Given an existing forecast, should a user retain all of its estimated intercept component, remove it, or attenuate it? Removing a reliably estimated component loses information, whereas retaining an imprecise one can increase forecast error. I calibrate its amplitude using previously issued forecasts and their completed outcomes, and compare the resulting rule with whole-forecast scaling, a characteristic-principal-component decomposition and forecast averaging. Over the matched 93-month horizon, residual calibration beats the complete forecast in 20 of 24 specifications, but beats averaging in only 8 and the principal-component control in 6. Under VW training and evaluation, gains over both controls reverse sign between expanding and rolling training windows for every factor count. The evidence shows why improvement over an unadjusted forecast is insufficient to establish the value of a particular factor-space decomposition.

The intended population matters for each decision. Equal-stock loss assigns as much importance to an error on a small stock as on a large one; capitalization-weighted loss emphasizes the latter. In this history, the VW-trained model predicts larger stocks better, whereas EW training is more competitive on micro and nano stocks under equal-stock loss.

Group accounting shows how those differences combine into aggregate performance. This links the evaluation to a model user’s coverage objective without translating forecast accuracy into a trading payoff. The common prediction target is clipped next-month excess return; the design and Appendix I explain the return treatment and the outstanding source-data and inferential limits.

The contribution is a matched empirical assessment of three questions that a residual comparison can otherwise conflate: what explains its magnitude, whether it helps select a forecast, and whether its component is useful within that forecast. The characteristic-factor setting follows Kelly et al. (2019). Pricing-error specification and testing are studied by Zhang (2024), while Choi and Yuan (2025) develop estimation and inference for inside and outside alphas. The present projection controls support an empirical comparison rather than a new pricing-error definition or test. The representation identities, also studied in Liu (2026), make the objects comparable; their contribution here is to identify what a reported difference changes.

The forecasting controls also locate the result within established estimation practice. Shrinkage of large return cross sections is studied by Kozak et al. (2020); the direct ridge rules used here are prediction benchmarks. Pástor and Stambaugh (2000) study investment decisions under uncertainty about pricing models, and Pelger and Xiong (2022) study state-dependent factor models. Against this background, the useful evidence is the specific crossing and the matched decision outcomes, including their reversals. The history is exploratory, and overlapping specifications are sensitivity comparisons rather than independent replications.

Section 2 defines the fitted signal and its residual. Section 3 describes the common forecasting task. Sections 4–6 examine interpretation, selection and component use. Section 7 explains why the measurement and estimation controls matter. The appendices follow the main text and provide derivations, detailed results and evidence limits.

2 From Pricing Errors to Fitted-Signal Comparisons

2.1 The economic benchmark and the estimated object

To interpret a pricing error, first fix what the pricing model is asked to explain. At formation month t , let N_t be the number of stocks in the specified population. Let $\mu_t \in \mathbb{R}^{N_t}$ be their population expected next-month excess returns conditional on a specified formation information set. A benchmark with K factors consists of exposures $\beta_t \in \mathbb{R}^{N_t \times K}$ and risk prices $\lambda_t \in \mathbb{R}^K$. Exposures are dimensionless and risk prices have monthly return units. With these objects and the return horizon fixed, the benchmark-relative pricing-error vector

is

$$\alpha_t = \mu_t - \beta_t \lambda_t \in \mathbb{R}^{N_t}. \quad (1)$$

The entries of α_t have monthly return units. They describe expected returns left unexplained by the specified benchmark, rather than unpredictable realized returns or proven market mispricing. For a general factor-pricing model, risk prices need not equal factor means. That equality requires the appropriate factor-return benchmark.

In the empirical analysis, both the mean signal and the factor structure are estimated through characteristics. The object studied is the characteristic-intercept contribution to a fitted mean under a stated factor-location convention. It represents a particular estimated pricing-error component; it does not by itself recover the full population vector in (1). Moreover, the empirical prediction outcome is a clipped excess return, so its fitted component refers to that transformed target.

Let $Z_t \in \mathbb{R}^{N_t \times L}$ collect the stocks' L dimensionless ranked characteristics. Its row $z'_{i,t}$ describes stock i . Let $y_{t+1} \in \mathbb{R}^{N_t}$ contain the empirical next-month return outcome in monthly decimal units, with clipping defined in Section 3. For a K -factor fitted representation, write

$$y_{t+1} = Z_t h_{t+1} + e_{t+1}, \quad h_{t+1} = B f_{t+1} + a. \quad (2)$$

Here $B \in \mathbb{R}^{L \times K}$ is the full-column-rank loading-coefficient matrix with dimensionless entries, $f_{t+1} \in \mathbb{R}^K$ is the latent factor vector in monthly return units, $a \in \mathbb{R}^L$ is an unrestricted, time-constant characteristic intercept, and $e_{t+1} \in \mathbb{R}^{N_t}$ is the realized residual vector. Both a and e_{t+1} have monthly return units. The complete coefficient $h_{t+1} \in \mathbb{R}^L$ maps characteristics to the fitted stock returns. Realized training factors can explain returns after they occur; a forecast must instead use an estimate of their mean available beforehand. The fitted equation defines the realized residual and does not impose a zero conditional mean on it. Throughout this section, unhatted coefficients describe a fixed fit; origin-specific estimates receive hats when the forecasting design is introduced.

An attribution within a fitted model must be distinguished from a new fit. For any transfer $d \in \mathbb{R}^K$, replacing a by $a - Bd$ and f_{t+1} by $f_{t+1} + d$ preserves every h_{t+1} . It preserves the entire fitted residual array, provided the observations, blocks and weights are unchanged. It also preserves a complete mean forecast if the factor-mean estimate receives the same transfer. A change in the factor-location convention can therefore redistribute a fixed fitted mean between its factor and intercept components. This leaves the complete prediction unchanged. Removing the intercept or changing the training loss instead changes the estimation problem.

2.2 Holding the signal and the measurement rule fixed

To compare two coefficient vectors on an economically specified population, let $D_t \in \mathbb{R}^{N_t \times N_t}$ be a diagonal matrix of nonnegative stock weights summing to one. Let $\mathcal{T}$ be a finite set of measurement months and let $v_t \geq 0$, with $\sum_{t \in \mathcal{T}} v_t = 1$, be its date weights. The characteristic Gram matrix and the signal norm are

$$Q = \sum_{t \in \mathcal{T}} v_t Z_t' D_t Z_t \in \mathbb{R}^{L \times L}, \quad \|b\|_Q^2 = b' Q b = \sum_{t \in \mathcal{T}} v_t \sum_i D_{t,ii} (z_{i,t}' b)^2, \quad (3)$$

where $b \in \mathbb{R}^L$ is a coefficient discrepancy in monthly return units. The final equality supplies the economic interpretation: two coefficient vectors differ to the extent that they imply different returns for the stocks being measured. The Gram matrix Q is dimensionless, and $b' Q b$ has squared monthly return units. In the projection results, assume Q is positive definite and B has rank K ; the empirical calculations check these conditions.

For example, two strongly correlated characteristics may receive opposite coefficients yet generate a small net signal across stocks. The Gram norm recognizes this cancellation. Equal-stock measurement gives each stock the same weight; capitalization-weighted measurement gives larger stocks more weight. These weights determine which stock signals matter most to the comparison. Moreover, Q contains characteristic second moments, not stock-return covariances. The quantity $b' Q b$ measures mean-signal magnitude and does not measure portfolio risk.

Now fix a target mean coefficient $g \in \mathbb{R}^L$, in monthly return units, and the loading space of B : the set of coefficients Bd that can be generated as the K -vector d varies. The question is how closely those fixed exposures can represent the target signal. For example, with one factor and two stocks whose exposures are one and two, factor-implied mean contributions must have the same one-to-two ratio. Changing the factor mean changes their level, but cannot generate every pair of stock means.

Allowing the factor-mean coordinates to be chosen for the specified target minimizes $(g - Bd)' Q (g - Bd)$ over $d \in \mathbb{R}^K$. The resulting $L \times L$ projection matrix $P_Q(B)$ and residual coefficient $r_Q(B, g) \in \mathbb{R}^L$ are

$$P_Q(B) = B(B' Q B)^{-1} B' Q, \quad r_Q(B, g) = [I_L - P_Q(B)]g, \quad (4)$$

where I_L is the L -dimensional identity matrix. The associated residual root mean square (RMS) in basis points is

$$\rho_Q(B, g) = 10^4 \sqrt{r_Q(B, g)' Q r_Q(B, g)}. \quad (5)$$

This is the smallest stock-weighted discrepancy between the specified mean signal and that loading space. The comparison lets each space use the factor-mean coordinates best suited to the target; it does not evaluate the original factor-mean forecast unchanged. Appendix A derives the minimizer and its orthogonality conditions. Holding g and Q fixed therefore gives loading spaces the same signal to explain under the same stock weights.

2.3 When the residual is the fitted characteristic intercept

For a model's own fitted mean, substitute $\bar{h} = a + B\bar{f} \in \mathbb{R}^L$, where $\bar{f} \in \mathbb{R}^K$ is the equal-date mean of training factors. To connect its residual to the intercept, let $d_Q = (B'QB)^{-1}B'Qa \in \mathbb{R}^K$ be a latent-factor location transfer, and define $a_Q = a - Bd_Q \in \mathbb{R}^L$ and $f_{t+1}^Q = f_{t+1} + d_Q \in \mathbb{R}^K$. Then

$$Bf_{t+1}^Q + a_Q = h_{t+1}, \quad B'Qa_Q = 0, \quad a_Q = [I_L - P_Q(B)]\bar{h}. \quad (6)$$

Thus $Z_t a_Q \in \mathbb{R}^{N_t}$ is the fitted characteristic-alpha component under this stated location convention. The operation assigns the part of the fitted mean outside the loading space to the intercept, while preserving every complete fitted return and mean forecast. Orthogonality holds under the historical measurement Gram; it need not hold in each month's cross section. The convention specifies the fitted factor-mean benchmark rather than establishing population risk prices. Appendix A supplies the derivation.

An external common target g yields a *spanning residual* $r_Q(B, g)$, which is not automatically that model's alpha. In the crossing experiment, diagonal entries compare each model with its own fitted mean; off-diagonal entries ask how well another space spans that mean. This naming makes the controlled comparison possible without assigning an external target's residual to the original model's intercept. Dividing the squared residual norm $r_Q(B, g)'Qr_Q(B, g)$ by $g'Qg$ yields a dimensionless residual share when $g'Qg > 0$. That share removes a nonzero scalar rescaling of the target, but not a change in its direction.

2.4 Representation and regularization

Holding the target fixed also requires a precise estimation rule. Consider an invertible characteristic recoding $\tilde{Z}_t = Z_t A$, where $A \in \mathbb{R}^{L \times L}$ is known and nonsingular. The same signal uses $\tilde{b} = A^{-1}b$ and the transported Gram matrix is $\tilde{Q} = A'QA$. Hence $\tilde{b}'\tilde{Q}\tilde{b} = b'Qb$. This recoding preserves the attainable linear signals; it adds no information about returns.

A regularized estimate need not be preserved if its penalty is left unchanged. For a quadratic coefficient penalty $b'Hb$, with $H \in \mathbb{R}^{L \times L}$ symmetric positive semidefinite, the equivalent penalty in the new coordinates is $\tilde{b}'(A'HA)\tilde{b}$. Using an identity-matrix ridge

penalty in both coordinate systems generally selects different signals. The baseline ridge controls therefore use one fixed formation encoding and training-only tuning. Section 7 then changes the encoding while explicitly comparing transported and new identity penalties. This experiment distinguishes a different estimate of the same information from an enlarged information set. Appendix B supplies the change-of-variables argument.

3 A Common Forecasting Task

Consider a research team reviewing two models that predict returns for the same stocks. Each model supplies both a complete expected-return signal and a factor structure used to interpret that signal. The team may compare their unexplained components, choose one complete forecast, or adjust a component within an existing forecast. These are three different decisions. The design first separates changes in the fitted signal from changes in its factor space, then evaluates model selection and component adjustment against subsequent returns. Keeping the stocks, information dates and loss function common makes those comparisons interpretable.

3.1 Data, formation information and weights

The application uses the existing U.S. stock panel associated with the characteristic library of Jensen et al. (2023). There are 153 characteristics and 749 formation months, July 1963–November 2025, with returns in the following month. The stored vintage determines upstream security eligibility and characteristic availability. The analysis rebuilds the characteristic encoding and common-population moments rather than combining exhibits from different data versions.

Within a formation month, each observed characteristic receives its average rank among nonmissing observations, divided by the number of nonmissing observations plus one, minus one half. Missing values receive zero. Ranks are computed before filtering on next-month return availability. Formation-market-capitalization weights use positive, available capitalization at that date. Estimation and forward loss share the subset with both a finite subsequent return and positive formation capitalization. Equal-stock weighting (EW) gives every included stock the same within-month importance; value weighting (VW) assigns importance in proportion to its formation capitalization. Training weights determine which fitting errors matter most when a model is estimated. Evaluation weights determine which subsequent forecast errors matter most when the two models are compared. Thus a VW-trained model can be evaluated with EW weights. Throughout, these choices change weights within a

common stock subset and characteristic encoding; they do not denote traded portfolios.

The dependent variable is next-month excess return clipped at the monthly first and ninety-ninth percentiles of all finite-return formation stocks, before imposing the common capitalization subset. This treatment limits the influence of extreme recorded returns and defines the prediction target throughout the paper. It is not the return on a traded portfolio. Independent reconstruction of the underlying return vintage and delisting treatment is incomplete; Appendix I documents the source checks and their limits. Formation-only Gram matrices, used for forward diagnostics, do not condition on subsequent-return availability. These matrices and the observed-return moments are deliberately kept distinct.

The estimation objective can be evaluated from monthly characteristic and return moments. Let $s \in \{\text{EW}, \text{VW}\}$ index training weights, let D_t^s contain the corresponding normalized stock weights, and write $W_t^s = Z_t' D_t^s Z_t \in \mathbb{R}^{L \times L}$, $X_t^s = Z_t' D_t^s y_{t+1} \in \mathbb{R}^L$ and $Y_{2,t}^s = y_{t+1}' D_t^s y_{t+1} \in \mathbb{R}$ for the moments on the common observed-return subset. The matrix W_t^s is dimensionless, X_t^s has monthly-return units and $Y_{2,t}^s$ has squared-return units. In this paragraph N_t counts that subset. For a training set $\mathcal{T}_o$ available at origin o , the scalar date weight is $p_{t,o} = N_t / \sum_{u \in \mathcal{T}_o} N_u$. The factor estimates minimize

$$\sum_{t \in \mathcal{T}_o} p_{t,o} [Y_{2,t}^s - 2(X_t^s)' h_{t+1}^s + (h_{t+1}^s)' W_t^s h_{t+1}^s], \quad h_{t+1}^s = B^s f_{t+1}^s + a^s. \quad (7)$$

The bracketed expression is the stock-weighted mean squared fitting error in month t . Weighting dates by their common stock counts keeps the EW/VW training comparison from also changing the relative importance of calendar months. The fitted return coefficient h_{t+1}^s contains that month's realized factor estimate. To turn it into a forecast, the primary specification replaces realized factors by their equal-date training average; the stock-count-weighted factor average is a separate sensitivity. Estimation date weights and factor-averaging weights therefore have distinct roles.

3.2 Estimation and the information sequence

The full-sample crossing uses three factors and five shared loading initializations, retaining the lowest attained criterion. The annual experiment considers $K = 1, 3, 5$, expanding training and a rolling 240-month window. Its first origin includes formation observations through February 2017 and their returns through March 2017. The first forecast uses March formation characteristics to predict April returns. Subsequent origins update annually through March 2025. This produces 105 target months, April 2017–December 2025: eight twelve-month blocks and a final nine-month block. For example, the 2017 decision uses completed returns

through March 2017 before forecasting April 2017. Appendix C records the corresponding formation-date convention in the replication.

At each annual origin, the full estimator updates its loading space, intercept and factor history. Multiple starts include singular-vector initialization and, after the first origin, the previous loading estimate and a fixed-space feasible fit. Three-factor expanding fits also compete with the earlier five-start estimates on the identical training observations. The selected full criterion is checked against the fixed-space feasible criterion. All retained starts converge under the recorded parameter and criterion tolerances. These checks limit optimization artifacts without proving global optimality.

For each model j , let $\hat{B}_{j,o} \in \mathbb{R}^{L \times K}$, $\hat{a}_{j,o} \in \mathbb{R}^L$ and $\hat{f}_{j,o} \in \mathbb{R}^K$ be its estimates at origin o . The complete mean coefficient and stock forecast are

$$\hat{h}_{j,o} = \hat{a}_{j,o} + \hat{B}_{j,o} \hat{f}_{j,o}, \quad \hat{y}_{j,i,t+1} = z'_{i,t} \hat{h}_{j,o}. \quad (8)$$

The coefficient remains fixed until the next annual origin, while stock characteristics change monthly. Thus two stocks can receive different forecasts, and a stock's forecast can change during the year, even without re-estimating the coefficient. The ordering is feasible within the stored data: training outcomes precede forecast outcomes. Because the historical period has already informed the research design, the comparisons remain exploratory historical evidence.

3.3 Diagnostics available before the next forecast block

The primary pair consists of the full EW-trained and VW-trained three-factor models with expanding estimation. To compare their residuals on the stocks relevant to a declared task, use the evaluation weighting $q \in \{\text{EW}, \text{VW}\}$. At origin o , let $Q_o^q \in \mathbb{R}^{L \times L}$ be the equal-date average of the preceding 60 available formation-only Gram matrices under those weights, ending at the last training formation month. Both candidates face the same matrix. The notation suppresses the superscript q and uses Q_o whenever the evaluation task is fixed.

The first diagnostic asks how much of each model's own fitted mean lies outside its loading space. This *own-mean diagnostic* evaluates $\rho_{Q_o}(\hat{B}_{j,o}, \hat{h}_{j,o})$. A second, *common-target diagnostic*, substitutes the same historically estimated mean coefficient $g_o \in \mathbb{R}^L$ for both candidates. It asks which space better represents one fixed reference signal, rather than comparing two spaces against different fitted means. The primary reference uses pooled ridge regression under the evaluation weights, trained on $\mathcal{T}_o$. This regression estimates one complete mean directly from characteristics and returns. Let $\bar{W}_o = \sum_t p_{t,o} W_t$ and

$\bar{X}_o = \sum_t p_{t,o} X_t$ denote its $L \times L$ Gram and L -vector of return moments. Then

$$g_o = (\bar{W}_o + \lambda_o c_o I_L)^{-1} \bar{X}_o, \quad c_o = \text{tr}(\bar{W}_o)/L. \quad (9)$$

Here $\lambda_o \geq 0$ is a dimensionless ridge parameter and c_o scales the penalty to the average characteristic second moment. The last 60 training returns choose λ_o from $\{0, 0.01, 0.1, 1, 10\}$ using coefficients fitted on earlier training months; the chosen value is then refitted using all training months. Using this same target makes the spanning comparison common to both spaces. The reference is nevertheless estimated from historical data shared with the factor models; its accuracy is not established by using it in both comparisons.

For either diagnostic, the proposed selection rule issues the complete forecast of the model with the smaller residual before the next block. This is a testable decision rule: good representation of a historical mean need not imply smaller future errors from that model's complete forecast. Numerical ties use an equal forecast average. Sensitivities replace the target by a pooled unpenalized mean or the preceding origin's ridge mean, replace the ruler by a training-window Gram, and normalize each own residual by its own mean-signal norm. Full versus fixed-space updating supplies an additional candidate pair. The first-origin full/fixed comparison is a structural tie and is excluded from ranking accuracy.

3.4 Loss, benchmarks and the unit of evidence

Forecast quality is evaluated by how closely the issued stock signals predict subsequent outcomes. For the common evaluation weighting $q \in \{\text{EW}, \text{VW}\}$, monthly mean squared error has the exact moment representation

$$\ell_{j,t+1}^q = Y_{2,t}^q - 2(X_t^q)' \hat{h}_{j,o} + \hat{h}_{j,o}' W_t^q \hat{h}_{j,o}. \quad (10)$$

This is the stock-weighted average of squared forecast errors, expressed using the same monthly moments as the estimator. Its units are squared monthly returns. Smaller equal-month average loss over the next block defines that origin's subsequent winner. Over a common evaluation set $\mathcal{H}$ of target months, prediction R^2 is

$$R_{\text{pred}}^2 = 1 - \frac{\sum_{t+1 \in \mathcal{H}} \ell_{j,t+1}^q}{\sum_{t+1 \in \mathcal{H}} Y_{2,t}^q}. \quad (11)$$

The denominator is the loss of a zero-return forecast on the identical observations. A positive value therefore means that the issued forecast reduces squared prediction error relative to zero. Tables report levels in percent and differences in percentage points. Forecast averages

are evaluated after averaging the predicted returns, so their squared loss includes the cross term.

A selector is useful only if its changing choices improve on a simple available alternative. Always choosing either complete model and equally averaging their forecasts are therefore essential controls. Direct pooled ridge and the time mean of monthly ridge coefficients challenge the factor construction itself. Monthly ridge tuning also uses only training observations. Unregularized monthly regressions are shown only where all training Grams pass the rank check; 99 early months fail that condition in the expanding sample. No pseudoinverse is silently substituted for the unavailable inverse.

There are nine annual decisions, not millions of independent selection trials. I report choice frequencies, correct rankings, balanced accuracy, losses, and leave-one-origin-out sensitivity. Balanced accuracy averages the correct-selection rates in the two realized winner classes; a rule that always chooses one candidate scores one half when both classes occur. Related specification cells share observations. These descriptive summaries are not full-procedure confidence intervals. Appendices C and I give timing tests, complete-year comparisons, ordinary predictive controls and the scope of the uncertainty statements.

4 The Mean behind the Residual Gap

The first experiment examines a familiar model-review argument: a new model leaves less of its estimated expected return unexplained, so its factor structure appears better. That comparison is incomplete when both the expected-return estimate and the factor structure have changed. The experiment therefore separates the two changes before assessing any forecast choice.

Estimate the model twice on the same stocks, giving either equal importance to each stock (EW training) or more importance to larger stocks (VW training). Each fit produces a complete mean signal and a loading space. Table 1 pairs each of the two signals with each of the two spaces. Within a panel, the same measurement Gram converts all four residuals into monthly basis points. The factor models are held fixed; each pairing is projected anew to find how closely the selected space can represent the selected mean. Thus the off-diagonal cells assess spanning ability, rather than the factor forecast originally issued by either model.

Table 1: Crossing the fitted mean with the factor space

Loading space estimated by	Fitted mean: equal-stock training	Fitted mean: cap-weight training
<i>Panel A: equal-stock measurement</i>		
Equal-stock training	117.37	75.02
Cap-weight training	116.67	77.27
<i>Panel B: capitalization-weighted measurement</i>		
Equal-stock training	93.75	63.12
Cap-weight training	92.49	63.52

Each cell measures the part of the column's fitted mean that the row's loading space cannot represent. Entries are monthly basis points (bp) of residual root mean square (RMS): $10^4 \sqrt{r^\top Q r}$, where $r \in \mathbb{R}^{153}$ is the projection residual coefficient and $Q \in \mathbb{R}^{153 \times 153}$ is the measurement Gram. Equal-stock (EW) training gives every stock equal influence within a month; capitalization-weighted (VW) training weights stocks by formation market capitalization. Both models use the same stocks, 153 characteristics, three factors and an unrestricted characteristic intercept, with July 1963–November 2025 formation dates. Within each panel, Q is fixed at the equal-month average over observed-return stocks. Each pairing is projected anew without re-estimating either model. Reading down a column changes only the space; reading across a row changes only the fitted mean. These full-sample residuals describe fitted clipped-return signals.

Start with the EW-measurement panel. Comparing the two diagonal cells changes both objects: the residual falls from 117.37 to 77.27 bp. Holding the EW mean fixed and changing only the space instead gives 117.37 versus 116.67 bp, a reduction of only 0.70 bp. Holding the EW space fixed and substituting the VW mean gives 117.37 versus 75.02 bp. The new mean is much closer even to the old space. Moreover, for the VW mean, the EW space leaves a slightly smaller residual than the VW space: 75.02 versus 77.27 bp. The large diagonal gap therefore chiefly accompanies a change in the signal being explained.

The VW-measurement panel reaches the same conclusion on a capitalization-weighted population. Its diagonal residuals are 93.75 and 63.52 bp; changing the space while holding the EW mean fixed only reduces 93.75 to 92.49 bp. A symmetric decomposition of the squared diagonal gap averages the two possible paths through the four cells. It attributes approximately 101% to the mean change and -1% to the space change under EW measurement, and 98% and 2% under VW measurement. Appendix B gives the decomposition. The negative share means that the space change offsets a small part of the mean contribution; these shares describe squared signal sizes, not percentages of the linear basis-point difference or causal effects.

This finding has a concrete economic source. Capitalization-weighted training places greater importance on fitting large-stock returns; equal-weight training gives each included

stock the same influence within a month. The resulting mean estimates can differ substantially even when the two factor spaces represent a common signal similarly. A researcher evaluating a replacement forecast should therefore ask whether a reported reduction in its estimated pricing-error component reflects greater explanatory capacity for a fixed signal or a change in the signal itself. Only the diagonal cells correspond to each model’s characteristic intercept under the stated location convention; the common name for all four cells is fitted-mean projection residual.

The crossing establishes the source of this observed residual contrast, without identifying which fitted mean is closer to the unknown conditional mean. Its mean change can involve direction as well as amplitude, so it is not identified as scalar shrinkage. The next question is empirical: after holding a reference mean fixed, does a space’s smaller historical residual help choose the better complete forecast? Before testing that rule, the analysis identifies the stocks contributing to the residual and distinguishes their fitted-signal discrepancies from their subsequent forecast errors.

4.1 Whose unexplained signal is being measured?

A residual norm aggregates discrepancies across stocks. Its economic meaning therefore depends on which stocks contribute most. A team screening a broad stock universe and a team concentrating on larger stocks may care about different parts of that aggregate. Figure 1 partitions each fitted mean’s squared residual by the stored formation size groups, keeping its residual coefficient and the original stock weights fixed. This locates a given signal discrepancy without fitting a new model for each group.

For group c , let $Q_c \in \mathbb{R}^{L \times L}$ be its contribution to the full measurement Gram, retaining the original population weights, and let $m_c \in [0, 1]$ be its average weight mass. The matrices satisfy $\sum_c Q_c = Q$. For a fixed residual coefficient $r \in \mathbb{R}^L$, its contribution and its within-group RMS are

$$s_c = \frac{r'Q_cr}{r'Qr}, \quad \rho_c = 10^4 \sqrt{\frac{r'Q_cr}{m_c}}, \quad m_c > 0. \quad (12)$$

For $r'Qr > 0$, s_c is a dimensionless share; ρ_c is in monthly bp and requires $m_c > 0$. The shares add to one because the same nonzero residual is used in every group. If the total residual is zero, its shares are undefined and there is no residual mass to allocate. Re-estimating a separate projection within each group would answer a different question and would not provide this accounting.

Under EW measurement, micro and nano stocks receive 58.0% of the population weight but account for 68.2% of the EW model’s squared residual. Their within-group residual RMS values are 123.5 and 134.8 bp, compared with 93.1 bp for mega stocks. The VW model’s

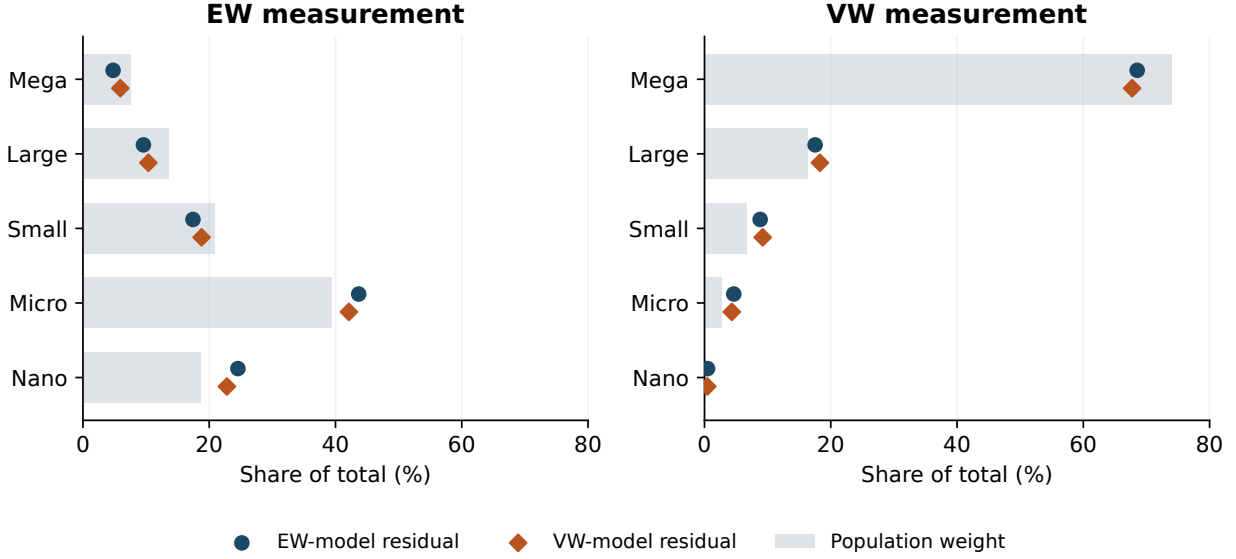


Figure 1: Where the residual signal is concentrated.

Gray bars show each size group's population weight; markers show its share of the squared residual norm for the EW-trained and VW-trained models. EW training weights stocks equally, while VW training uses formation capitalization. Each panel uses one fixed full-population measurement Gram; each model's residual is projected once and then partitioned across groups. Formation dates are July 1963–November 2025. A marker above the corresponding bar identifies a disproportionate contribution to the fitted residual, measured as an additive share rather than within-group RMS. The unclassified group has zero mass. The comparison locates the residual signal; Table 2 separately evaluates subsequent forecast performance within these populations.

residual is smaller throughout these groups, rather than only in the smallest stocks. Under VW measurement, mega stocks account for most of the population weight and residual mass. The concentration figure therefore identifies whose fitted signal contributes to the aggregate residual. Its interpretation is exposure to a mean-signal discrepancy on those stocks, rather than their portfolio risk or the profitability of trading the residual.

To check whether the full-sample attribution also appears in the information available to a forecaster, repeat the four-cell crossing at each annual origin. The comparison attributes between 96.8% and 100.9% of the squared diagonal gap to the mean change under EW measurement, and between 95.5% and 100.8% under VW measurement. Table E.4 reports every origin. The same attribution therefore recurs in the historical fitted signals available before the next forecast block, rather than appearing only in a fit that uses the full return history.

4.2 Population-specific forecasting consequences

Residual concentration alone does not tell the user how well a model predicts the stocks carrying that residual. Table 2 therefore turns to subsequent outcomes, keeping the same annually issued forecasts and evaluating them separately on larger stocks, micro stocks and nano stocks. The larger group combines the stored mega, large and small labels. The experiment changes only the evaluation population: models, formation ranks, clipping thresholds and annual forecast dates stay fixed. Within each group and month, weights are renormalized to one so that prediction R^2 describes that group’s forecasting task. The comparison can therefore reveal heterogeneity hidden by the whole-population loss.

Table 2: Forecast performance differs across stock populations

Population	EW-trained	VW-trained	Average	Ridge
<i>Equal-stock evaluation: within-group prediction R^2 (%)</i>				
Larger stocks	-0.0698	0.1681	0.1419	0.1004
Micro	0.1289	0.0725	0.1638	0.1780
Nano	0.9010	0.7773	0.8977	0.8156
<i>Cap-weight evaluation: within-group prediction R^2 (%)</i>				
Larger stocks	0.5850	1.0809	0.9844	0.8780
Micro	0.0092	0.0354	0.0827	0.1417
Nano	0.8396	0.6364	0.8010	0.5660

Larger stocks combine the source data’s mega, large and small size groups; micro and nano are separate groups. Entries evaluate the same three-factor expanding forecasts as Table 3, now within each stock group, over 105 months from April 2017 to December 2025. EW-trained and VW-trained refer to equal-stock and formation-capitalization training. Evaluation weights are renormalized within each month and group before accumulating squared losses and the zero-forecast denominator. These within-group R^2 levels therefore do not average to aggregate R^2 . Average equally combines the complete EW/VW forecasts. Ridge is fitted to the full evaluation population without group-specific retuning. Characteristic ranks and clipping cutoffs remain those of the full common population. No observations are unclassified. Tables E.2 and E.1 report group masses, selector outcomes and additive contributions to aggregate performance.

Under EW evaluation, the VW-trained forecast improves on the EW-trained forecast for larger stocks, with R^2 of 0.1681% against -0.0698% . The ordering reverses among micro stocks, 0.0725% against 0.1289%, and nano stocks, 0.7773% against 0.9010%. Under VW evaluation, the larger-stock group carries 99.0% of the average stock-weight mass in the forecast period. Its ranking consequently dominates the aggregate capitalization task. The same pair of forecasts can consequently receive different rankings depending on the population being predicted. This is a performance comparison within recorded groups; changing a stock’s capitalization is not an intervention in the experiment.

An additive loss decomposition makes the aggregate trade-off explicit. Under EW evaluation, replacing the EW forecast by the VW forecast gains 0.06755 percentage points from larger stocks, loses 0.02761 from micro stocks and loses 0.02160 from nano stocks. Using unrounded contributions, the net gain is 0.0183 points. Conditional group R^2 values themselves cannot be averaged to obtain this number, because their denominators differ; the appendix uses the common full-population denominator. For a model user, the lesson is to specify the intended stock population before choosing a model, then inspect which groups produce its aggregate advantage. A smaller residual in a model’s own fitted mean does not supply this population-specific evidence about its future prediction errors.

5 Do Residual Diagnostics Anticipate Forecast Rankings?

5.1 A diagnostic choice and its subsequent outcome

The crossing explains why a smaller residual need not reflect better representation of a fixed mean. The next question is whether the residual provides useful information for a different decision: which complete forecast should a user issue next? At each annual origin, the selection rule chooses the candidate with the smaller historical residual and is then judged on the next block’s returns. The candidate forecasts, stock population and outcome dates are held fixed; only the selection rule changes. The comparison tests the diagnostic’s decision value, rather than a null hypothesis that population pricing errors are zero.

Begin with the residual of each model’s own fitted mean. In the primary three-factor expanding specification, the own-mean diagnostic selects the VW-trained model at all nine origins under either evaluation weighting. It correctly selects five of nine subsequent winners under EW loss and six of nine under VW loss. Since both choices are constant, their balanced accuracies are 50%. These hit rates provide no incremental selection information beyond always using the VW-trained model.

Normalizing the own residual by its own mean-signal norm leaves these primary choices unchanged. The finding is therefore not removed by an amplitude-free version of the same diagnostic. The VW model can still be a useful forecast: it performs strongly under VW evaluation. The diagnostic, however, reproduces a fixed choice and therefore contributes no changing information to that decision.

Next hold the reference mean fixed across the two spaces. The common ridge target changes the selections. Under EW evaluation it chooses the EW-trained model seven times and the VW-trained model twice; four of its nine choices are correct. Under VW evaluation

it chooses the EW-trained model once and the VW-trained model eight times; five choices are correct. Balanced accuracies are 47.5% and 41.7%, respectively. Thus holding the target fixed can change the apparent relative strength of the spaces without yielding a more useful ordering for future complete-forecast loss.

Figure 2 goes beyond correct-choice counts by comparing the size of the available diagnostic difference with the size of the subsequent forecast difference. Its horizontal coordinate is the EW residual RMS minus the VW residual RMS, both evaluated against the same historical ridge target. Its vertical coordinate is next-block VW prediction R^2 minus EW prediction R^2 . Positive values on either axis favor VW; points with the same sign on both axes indicate a correct diagnostic ordering.

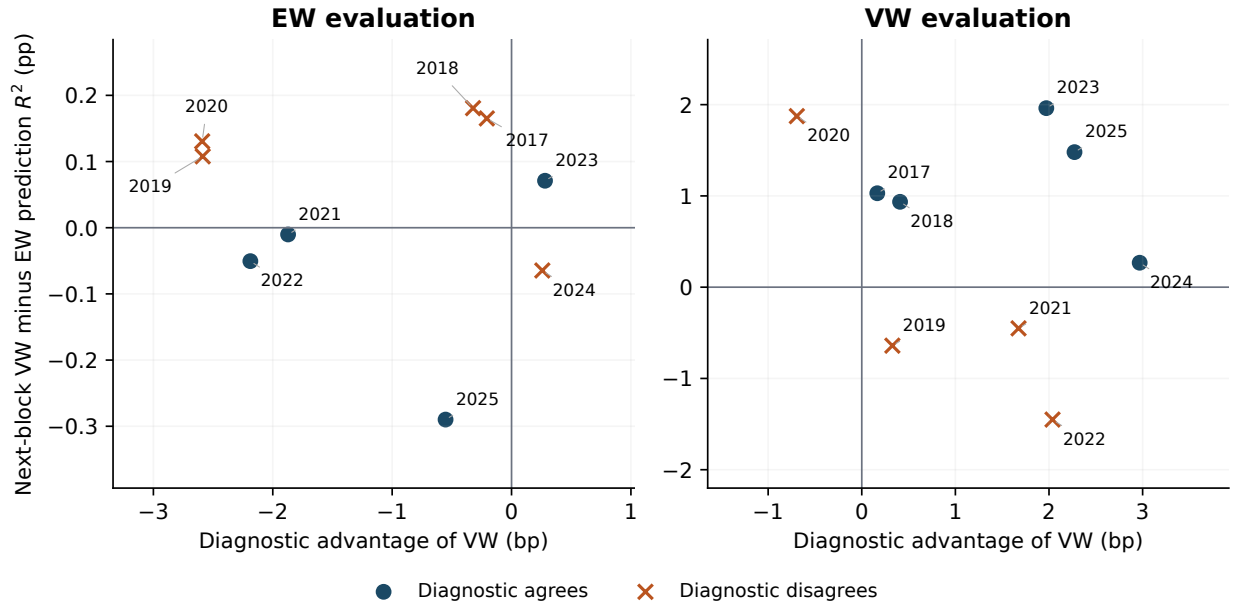


Figure 2: Diagnostic strength and subsequent forecast differences.

Each labeled point is one annual origin for the three-factor expanding models. The horizontal coordinate is the estimated EW-space minus VW-space residual RMS against the same historical ridge mean, in monthly basis points; a positive value favors the VW space. The vertical coordinate is subsequent VW-trained minus EW-trained prediction R^2 , in percentage points; a positive value favors the VW forecast. Matching signs in the upper-right and lower-left quadrants indicate a correct ordering. Both spaces use the panel's preceding-60-month formation Gram. EW means equal-stock weights and VW formation-capitalization weights. The 2025 block has nine target months and the others twelve. Different vertical scales are used across panels. The scatter shows whether a stronger historical spanning advantage corresponds to a larger later forecasting advantage.

For instance, the VW-evaluation diagnostic favors VW at the 2022 origin by about 2.04 bp, yet that model subsequently trails by 1.45 percentage points. At the 2023 origin a similar diagnostic advantage, 1.97 bp, precedes a gain of 1.96 points. In these observations,

a similar historical spanning advantage can therefore precede opposite forecast outcomes. The distinction matters because the selected model issues its own complete mean; a good projection of the common ridge reference does not guarantee the accuracy of that issued mean. All nine origins remain visible, including the shorter 2025 block. Table C.1 and Figure C.1 give the corresponding categorical choices.

5.2 The loss of following the diagnostic

Ranking accuracy does not measure the size of forecasting mistakes. Table 3 therefore evaluates the complete forecast issued by each rule over the same 105 months. Under EW evaluation, the common-ridge selector produces prediction R^2 of 0.3612%, below the equal forecast average at 0.4396% and direct pooled ridge at 0.3972%. Under VW evaluation, the corresponding values are 0.7326%, 0.9569% and 0.8550%. The VW-trained component itself reaches 1.0491% in the VW task. Forecast averaging is a demanding simple control, but these results do not make it a universal winner.

Table 3: Complete-forecast performance on a common horizon

Forecast rule <i>Prediction R^2 (%)</i>	EW evaluation	VW evaluation
EW-trained factor model	0.3605	0.5678
VW-trained factor model	0.3788	1.0491
Own fitted-mean selector	0.3788	1.0491
Common-ridge selector	0.3612	0.7326
Equal average: EW and VW factors	0.4396	0.9569
Direct pooled ridge	0.3972	0.8550
Mean of monthly ridge regressions	0.3112	1.0435

Prediction R^2 is 100 times one minus total weighted squared forecast error divided by the corresponding zero-return forecast loss. Thus entries are percent levels, and larger values indicate more accurate forecasts. All rows use the same 105 target months, April 2017–December 2025, and the same observed-return stocks. EW evaluation gives stocks equal weight; VW evaluation uses formation market capitalization. Factor models have three factors, expanding estimation and unrestricted characteristic intercepts; complete means use equal-date factor averages. The own-mean selector compares residuals of each model's own fitted mean, while the common-ridge selector holds the historical reference mean fixed across spaces. Both use the preceding-60-month formation Gram. Ridge tuning uses training observations only. The equal average combines both complete factor forecasts, with their cross term retained in the loss. Outcomes are monthly excess returns clipped at common first and ninety-ninth percentile cutoffs.

Figure C.2 accumulates the selector's paired loss relative to the equal average. This shows whether the terminal gap emerges gradually or is dominated by particular parts of

the history. Removing any one annual block leaves the common-target selector below the average in prediction R^2 under both evaluation weights. The full-year comparison, which removes the last nine-month block, also preserves that ordering. The observed ranking is therefore not attributable to a single annual block; the deletion exercise measures this sensitivity without providing a confidence interval for another sample.

A common target removes one source of incomparability, but the historical period used to measure its residual remains a design choice. Replacing the preceding-60-month Gram by the entire training-window Gram raises the EW common-target hit rate from four to six of nine. Its prediction R^2 rises to 0.3847%, still below the equal average's 0.4396% and pooled ridge's 0.3972%. Thus a longer measurement window improves the selector in this cell, without reversing its comparison with these two simple benchmarks. Using the unpenalized pooled mean instead restores the constant preference for VW in the primary pair. Appendix C reports these reference changes together, so the favorable window is assessed alongside the others.

Across the twelve EW/VW decision sequences formed by three factor counts, two training windows and two evaluation weights, the own-mean diagnostic always chooses VW. The common-ridge selector exceeds the equal average in five sequences and direct pooled ridge in four. Additional full/fixed-space comparisons also give mixed results. These overlapping sequences define the robustness range of the experiment; they are not twelve or twenty-four independent studies. The common-target construction makes the diagnostic easier to interpret, but the tested rule does not consistently turn that interpretability into lower forecast loss than the declared controls.

For a practitioner, these results suggest evaluating a diagnostic against the decision it is intended to change. A residual measure can help explain a fitted signal, while a model-selection rule must also improve on an available fixed forecast or combination for the intended stocks. The reported loss comparison supplies that second assessment; an investment implementation would require the separate portfolio specification described in Appendix I.

Why is averaging a demanding control? Under squared loss, averaging two forecasts reduces the loss associated with their disagreement. To make the comparison exact, hold the evaluation month and weights fixed. For complete coefficient vectors $h_A, h_B \in \mathbb{R}^L$ and their average $h_C = (h_A + h_B)/2$, direct quadratic expansion gives

$$\frac{\ell(h_A) + \ell(h_B)}{2} - \ell(h_C) = \frac{1}{4}(h_A - h_B)'W_t(h_A - h_B). \quad (13)$$

The right side is one quarter of their stock-weighted squared signal disagreement. It is non-negative, so the average forecast beats the arithmetic mean of the two component losses,

equivalently the expected loss of equal-probability random selection. It need not beat either component separately. Across the 105 target months, this reduction contributes 0.0700 prediction- R^2 percentage points under EW evaluation and 0.1485 under VW evaluation, relative to the average component loss. A selector must use informative choices to compensate for giving up that reduction. A separate decision remains: given one complete forecast, can adjusting its residual component improve how it is used?

6 Using the Residual Signal within a Forecast

6.1 A forecast-component decision

Model selection asks which complete forecast to use. Component calibration asks how to use a forecast already selected. A research team may retain a model’s estimated factor structure while questioning how much of its remaining expected-return signal to include. That residual may contain useful prediction information, but its estimated amplitude may also be noisy. Deleting it sacrifices possible signal; retaining it fully preserves its estimation error as well. This trade-off motivates a separate calibration experiment, holding the fitted models and information dates fixed. Its rationale does not depend on the success or failure of the preceding selection rule.

For a model and origin, suppress their subscripts and write $p = P_{Q_o}(\hat{B})\hat{h} \in \mathbb{R}^L$ for the projected mean coefficient and $r = \hat{h} - p \in \mathbb{R}^L$ for its residual. Both are available at the origin and have monthly return units. Under the location convention in Section 2, r is the characteristic-intercept component of this model’s own fitted mean. Here p denotes a projected coefficient, distinct from the scalar date weights $p_{t,o}$ used in estimation. For a dimensionless retention fraction $\eta \in [0, 1]$, issue

$$h(\eta) = p + \eta r. \tag{14}$$

At $\eta = 0$, the forecast uses only the factor-space projection; at $\eta = 1$, it uses the complete mean. For example, $\eta = 1/2$ retains the projection and halves the residual contribution to every stock’s forecast. The operation changes the issued signal while preserving the estimated loading space. The measurement matrix is the evaluation population’s preceding-60-month formation Gram, so the component being scaled is defined for the stocks being predicted. The projected component is itself estimated; keeping its coefficient at one is a restriction that the controls below examine.

To determine what the residual must contribute to justify retaining it, hold the issued p

and r fixed and compare their subsequent squared losses. The exact loss expansion gives

$$\ell(p) - \ell(p + r) = 2r'(X_t - W_t p) - r'W_t r. \quad (15)$$

All moments use the same next-month evaluation observations. The first term measures alignment between the issued residual signal and the realized error of the projected forecast; the second is its squared signal size. Full retention improves the realized prediction only when that alignment exceeds the squared-size cost. A large estimated residual can be harmful if it does not point toward the projected forecast’s subsequent errors; a smaller but well-aligned residual can help. Historical orthogonality fixes the decomposition but does not determine this future alignment. Appendix F derives the identity and reports its sample accounting.

6.2 From a diagnostic to an issued forecast

As an initial diagnostic, evaluate the fixed fractions $\eta \in \{0, 1/4, 1/2, 3/4, 1\}$ on all 105 forecast months. Figure F.1 reports every fraction. An interior fraction beats both endpoints in each of the four training/evaluation weighting pairs, suggesting that partial retention is worth examining. This comparison uses the full realized history to describe the curve; it does not yet supply a fraction available before a forecast is issued.

Starting in 2018, estimate η from the preceding twelve completed forecast months, using the projected and residual coefficients actually issued at the previous origin. Constrain the scalar to $[0, 1]$ and apply it to the current residual. This yields eight annual calibration decisions and 93 target months, April 2018–December 2025. The shorter horizon gives every adaptive competitor a preceding forecast record before its first evaluation. Calibration uses the coefficients issued before those twelve outcomes, then applies the fitted scalar to the newly estimated components. It therefore estimates how much of the current residual to retain from a completed, historically available forecasting record.

Several controls distinguish the proposed interpretation from simpler explanations of a gain. The first three use exactly the same completed forecast record. Whole-mean calibration uses the same record to scale the entire forecast. A two-component rule estimates separate scales for its projection and residual, testing the restriction that the projected component is trusted at full strength. A characteristic-PC rule projects the same complete mean onto the leading K eigenvectors of the same formation Gram and calibrates its remaining component. It asks whether the estimated factor space adds value beyond a simple characteristic-space decomposition. Here PC means principal component of the characteristic second-moment matrix, not a component of stock-return covariance. This is a generic decomposition control; the relation to native pricing-error estimators is discussed in Appendix I. Appendix G states

the optimization problems.

An identity-matrix projection replaces stock-level signal distance by ordinary Euclidean distance in the original rank coordinates. It asks whether the Gram-based choice of residual matters once that component is actually scaled. Equal averaging, a past-calibrated convex combination, past-loss model selection and direct pooled ridge test whether adjusting one model is preferable to combining or replacing it. Adaptive component scales, the convex-combination weight and past-loss selection share the preceding twelve issued-forecast outcomes. Direct ridge and the common-target diagnostic retain their training-only estimation rules. All comparisons use the same information cutoff, outcomes and evaluation weights. In particular, the averaging benchmark uses complete EW and VW factor forecasts, rather than averaging reported R^2 values.

6.3 Calibration depends on the benchmark and training window

Table 4 reports the primary three-factor expanding comparison. Under EW evaluation, the equal forecast average has $R^2 = 0.4506\%$, above either residual-calibrated factor forecast. Under VW evaluation, residual calibration of the VW-trained model reaches 1.0962%, compared with 0.9689% for its complete forecast, 1.0109% for characteristic-PC calibration and 0.8876% for the average. The primary VW/VW cell therefore supports a useful component adjustment on that task, while the EW results show that adjusting a single model need not beat combining two complete forecasts.

Table 4: Residual calibration and competing forecast rules

Rule <i>Prediction R^2 (%)</i>	Equal-stock evaluation		Cap-weight evaluation	
	EW training	VW training	EW training	VW training
Complete forecast	0.3794	0.3886	0.5212	0.9689
Fixed half residual	0.4102	0.3696	0.7469	1.0849
Residual calibration	0.4181	0.3885	0.7405	1.0962
Whole-mean calibration	0.3814	0.3326	0.3372	-0.1165
Two-component calibration	0.3604	0.3324	0.2642	-0.0343
Characteristic-PC calibration	0.4346	0.3987	0.7616	1.0109
Identity-metric calibration	0.2595	0.2789	-0.0749	0.7795
Equal forecast average	0.4506	0.4506	0.8876	0.8876
Past convex combination	0.4532	0.4532	0.5953	0.5953
Pooled ridge	0.4055	0.4055	0.7517	0.7517
Past-loss selector	0.4208	0.4208	0.6502	0.6502
Residual-diagnostic selector	0.3801	0.3801	0.6331	0.6331

Entries are prediction- R^2 levels in percent; larger values indicate smaller squared forecast losses relative to the zero-return forecast. All rules use the same 93 target months, April 2018–December 2025, and eight annual calibration decisions. EW and VW training use equal stock and formation-capitalization weights, respectively; the column groups specify how subsequent stock-level losses are weighted. Factor models use three factors and expanding training. Component scales, convex-combination weights and past-loss selection use the preceding twelve completed outcomes of previously issued forecasts. Residual and whole-mean scales lie in $[0, 1]$; two-component calibration estimates a scale for each component. The PC control uses the leading three eigenvectors of the same preceding-60-month formation Gram; the identity-metric control projects in the original rank coordinates. Averaging and selection use both complete EW/VW forecasts, so their entries repeat across training columns. Ridge and residual-diagnostic selection use training-only estimates; ridge uses the evaluation weights. All comparisons share the clipped-return target and retain the forecast cross terms in squared loss.

Whether this favorable VW/VW result persists when the estimation history changes is the main robustness question. Figure 3 contrasts residual calibration with averaging and PC calibration for $K = 1, 3, 5$ under expanding and rolling 240-month estimation. In the VW/VW task, gains over averaging range from 0.1513 to 0.2086 percentage points with expanding training, but from -0.1554 to -0.0739 with rolling training. Gains over PC calibration also reverse sign between these windows for every factor count. Thus choosing a characteristic Gram is not sufficient to reproduce the expanding-window advantage. The useful comparison is between complete procedures: the window estimates the mean and space, and calibration then acts on their decomposition.

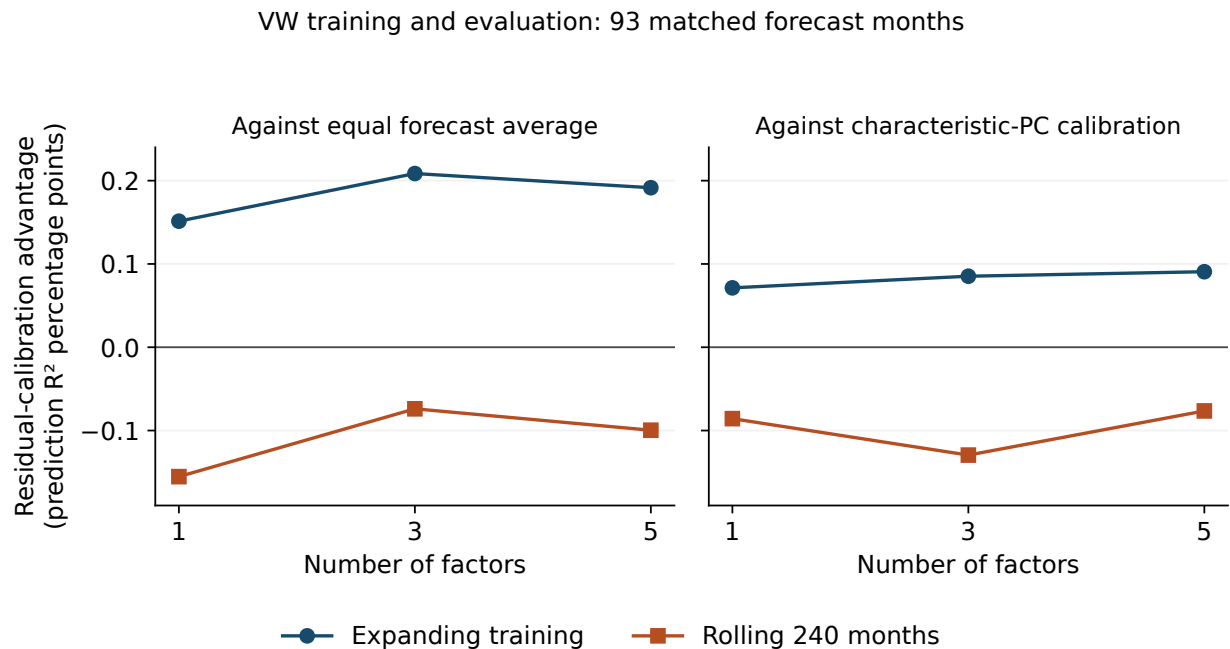


Figure 3: The calibration advantage reverses across training windows.

All models are trained and evaluated with formation-capitalization weights (VW). Points report factor-space residual calibration's prediction- R^2 gain, in percentage points, over the equal average of complete EW/VW forecasts (left) and characteristic-PC residual calibration (right). Positive values favor factor-space calibration. All rules use the same 93 target months, April 2018–December 2025, and eight annual calibration decisions. The PC control uses the leading K eigenvectors of the preceding-60-month formation Gram. Expanding windows use the available training history; rolling windows retain 240 months. Lines connect the three evaluated factor counts, without implying confidence bands. Gains are positive for every expanding specification and negative for every rolling specification: the calibration advantage depends on the training history. Table G.1 reports all training/evaluation pairs.

Across the complete grid, residual calibration exceeds full retention in 20 of 24 cells, whole-mean scaling in 20, and the two-component rule in 22. It exceeds PC calibration in

6 and equal averaging in 8. Table G.1 reports the complete grid. These cells share stocks, returns and training histories; the counts summarize specification sensitivity and are not independent statistical evidence.

The two-component control can perform poorly when twelve completed months lead it to reduce the projected component severely. Allowing more adaptation can therefore increase forecast error when the calibration record is short; the result does not establish that this component represents economically priced risk. For model use, the main lesson is to compare a proposed adjustment of the characteristic pricing-error component with ordinary shrinkage, simple decompositions and forecast combinations on the intended population. Its residual can contribute useful prediction information even when residual magnitude does not usefully rank models. The observed window reversals show why that contribution must be evaluated for the estimation and calibration procedure actually used.

7 The Role of Geometry in the Evaluation

The preceding experiments give geometry two distinct roles. The characteristic Gram translates a coefficient discrepancy into a stock-level signal discrepancy on the chosen population. It also defines which component is adjusted in the retention experiment. Changing the representation of a fixed complete forecast leaves its predictions unchanged; changing the weight on a component alters the forecast and must be evaluated on subsequent returns.

There is a further question behind the crossing result. With an unrestricted intercept, why can two factor fits produce different complete means at all? The intercept's first-order condition supplies the answer. Suppress origin and training superscripts, and hold the training observations and their weights fixed. Recall that $W_t = Z_t' D_t Z_t \in \mathbb{R}^{L \times L}$ is the monthly characteristic Gram and $X_t = Z_t' D_t y_{t+1} \in \mathbb{R}^L$ is the characteristic-return moment. Using the normalized training-date weights p_t , define $Q_p = \sum_t p_t W_t \in \mathbb{R}^{L \times L}$, $X_p = \sum_t p_t X_t \in \mathbb{R}^L$ and $f_p = \sum_t p_t f_{t+1} \in \mathbb{R}^K$. Assume Q_p is positive definite. Let $h_{\text{pool}} = Q_p^{-1} X_p \in \mathbb{R}^L$ denote the pooled regression coefficient. Let $\bar{f} \in \mathbb{R}^K$ be the equal-date mean of the training factors and $\bar{h} = a + B\bar{f} \in \mathbb{R}^L$ the complete fitted mean. Define the L -vector $C_{Wf} = \sum_t p_t (W_t - Q_p) B (f_{t+1} - f_p)$. It records sample co-movement between changing characteristic moments and realized factor contributions. At an unrestricted-intercept stationary fit,

$$\bar{h} = h_{\text{pool}} - Q_p^{-1} C_{Wf} + B(\bar{f} - f_p). \quad (16)$$

Appendix A derives the identity. The first term is the return signal from pooled regression. The second accounts for whether factor realizations coincide with months in which the cor-

responding characteristic exposures are unusually large or small. The last term reflects the use of different date weights to fit the model and to average its factors. Thus a free intercept need not make the complete factor-model mean coincide with the pooled mean.

If characteristic moments are constant, the co-movement term is zero. If the factor mean also uses the training-date weights, the final term is zero, and the complete mean equals pooled regression even with a fixed loading space. The general identity locates the estimation channels through which factor fits can produce different means. Whether those differences predict subsequent returns is the empirical question addressed by the forecasting controls.

Appendix H evaluates this distinction in three ways. Transporting both characteristics and their ridge penalty preserves the forecast; using a new identity penalty can change it. Full and fixed-space annual updates are compared with pooled and monthly ridge across the same factor-count and window grid. Finally, a controlled simulation crosses rotating characteristic moments with changing conditional factor means. Rotation alone does not create a predictable mean correction. Together, these controls connect the algebra to model use: a measured change in exposures, an altered estimator, and a useful forecast improvement are three different findings. The framework makes them separately testable.

8 Conclusion

A pricing-error comparison must specify what is held fixed. In this application, one estimator reports a much smaller fitted characteristic-intercept component, yet the two loading spaces represent a common fitted mean almost equally well. Crossing means and spaces attributes most of the contrast to the estimated signal. This tells a researcher what changed when a replacement model reports less unexplained return.

The forecasting exercises distinguish description from decision value. Historical residual magnitudes add little to model selection. Calibrating the residual often improves on full retention, but comparisons with principal-component calibration and forecast averaging depend on the training window. Group results also show why any ranking must specify its evaluation population.

The empirical contribution is this separation of interpretation, selection and component use on a common forecasting task. It suggests diagnosing the source of a residual contrast before acting on it, and testing each action against simple alternatives. The evidence concerns exploratory prediction of clipped U.S. stock returns; extending the conclusions requires source-return reconciliation, independent confirmation and suitable inference for the full estimation and decision procedure.

A Projection and Mean-Estimation Identities

A.1 Projection, transfer and the preserved signal

The first identity asks how closely fixed factor exposures can represent a chosen fitted mean. Fix a full-column-rank loading-coefficient matrix $B \in \mathbb{R}^{L \times K}$, a positive-definite characteristic measurement Gram $Q \in \mathbb{R}^{L \times L}$ and a mean coefficient $g \in \mathbb{R}^L$. The first two objects are dimensionless; g has monthly return units. The factor-mean coordinates $d \in \mathbb{R}^K$ are allowed to vary while these three objects remain fixed. The projection solves

$$\min_d (g - Bd)'Q(g - Bd). \quad (\text{A.1})$$

The gradient with respect to d is $-2B'Q(g - Bd)$, so the first-order condition is $B'QBd = B'Qg$. Since $B'QB$ is positive definite, the unique minimizer is $c_Q = (B'QB)^{-1}B'Qg \in \mathbb{R}^K$. The symbol c_Q denotes the coordinates for the complete target, distinct from the intercept transfer d_Q below. The residual $r = g - Bc_Q \in \mathbb{R}^L$ satisfies $B'Qr = B'Qg - B'QBc_Q = 0$. For any other d , expansion yields

$$(g - Bd)'Q(g - Bd) = \{r + B(c_Q - d)\}'Q\{r + B(c_Q - d)\} \quad (\text{A.2})$$

$$= r'Qr + 2(c_Q - d)'B'Qr + (c_Q - d)'B'QB(c_Q - d) \quad (\text{A.3})$$

$$= r'Qr + (c_Q - d)'B'QB(c_Q - d). \quad (\text{A.4})$$

The vanishing cross term proves the minimizing property and $g'Qg = (Bc_Q)'Q(Bc_Q) + r'Qr$. Financially, the residual is the part of the chosen mean signal that cannot be reproduced by those exposures under those stock weights. The proof makes no assertion about a true conditional mean or the best estimated loading space.

For the link to a model's fitted characteristic intercept, let $a \in \mathbb{R}^L$ be its intercept, $f_{t+1} \in \mathbb{R}^K$ its fitted factors, and $\bar{f} \in \mathbb{R}^K$ their equal-date training mean, all in monthly return units. Put $g = a + B\bar{f}$ and recall $P_Q(B) = B(B'QB)^{-1}B'Q$. Because $P_Q(B)B = B$, the residual is

$$(I_L - P_Q(B))(a + B\bar{f}) = (I_L - P_Q(B))a = a - B(B'QB)^{-1}B'Qa, \quad (\text{A.5})$$

where I_L is the $L \times L$ identity matrix. Define the intercept transfer $d_Q = (B'QB)^{-1}B'Qa \in \mathbb{R}^K$, the transferred intercept $a_Q = a - Bd_Q \in \mathbb{R}^L$, and transferred factors $f_{t+1}^Q = f_{t+1} + d_Q \in \mathbb{R}^K$.

$\mathbb{R}^K$. Direct substitution gives

$$Bf_{t+1}^Q + a_Q = Bf_{t+1} + Bd_Q + a - Bd_Q = Bf_{t+1} + a, \quad (\text{A.6})$$

$$B'Qa_Q = B'Qa - B'QBd_Q = 0. \quad (\text{A.7})$$

The fitted characteristic-alpha coefficient is therefore exactly the residual of this model's own mean under the stated convention. A researcher can change this attribution without changing the fitted returns or the complete mean forecast. A nonsingular factor rotation likewise preserves the complete index when loadings and factors are transported together. An intercept restriction, a finite residual budget, or a coefficient penalty can instead change the estimation problem and requires a separate analysis.

A.2 Complete means under an unrestricted intercept

This identity explains why an unrestricted intercept does not generally force the complete factor-model mean to equal a pooled regression mean. For training month t , let $Z_t \in \mathbb{R}^{N_t \times L}$ contain characteristics, $D_t \in \mathbb{R}^{N_t \times N_t}$ be diagonal nonnegative stock weights summing to one, and $y_{t+1} \in \mathbb{R}^{N_t}$ contain the next-month return outcome. Define $W_t = Z_t' D_t Z_t \in \mathbb{R}^{L \times L}$ and $X_t = Z_t' D_t y_{t+1} \in \mathbb{R}^L$. Hold these sample moments and the normalized training-date weights $p_t \geq 0$, $\sum_t p_t = 1$, fixed. Define $Q_p = \sum_t p_t W_t \in \mathbb{R}^{L \times L}$, $X_p = \sum_t p_t X_t \in \mathbb{R}^L$ and $f_p = \sum_t p_t f_{t+1} \in \mathbb{R}^K$. Assume Q_p is positive definite, and let $\bar{f} \in \mathbb{R}^K$ denote the equal-date factor mean. At a stationary fit with an unrestricted, unpenalized intercept, the derivative of the weighted squared-loss criterion with respect to a is zero. Dividing that derivative by -2 gives

$$0 = \sum_t p_t \{X_t - W_t(a + Bf_{t+1})\} = X_p - Q_p a - \sum_t p_t W_t B f_{t+1}. \quad (\text{A.8})$$

The centered geometry-factor product $C_{Wf} \in \mathbb{R}^L$ expands as

$$C_{Wf} = \sum_t p_t (W_t - Q_p) B (f_{t+1} - f_p) \quad (\text{A.9})$$

$$\begin{aligned} &= \sum_t p_t W_t B f_{t+1} - \left(\sum_t p_t W_t \right) B f_p \\ &\quad - Q_p B \left(\sum_t p_t f_{t+1} \right) + Q_p B f_p \end{aligned} \quad (\text{A.10})$$

$$= \sum_t p_t W_t B f_{t+1} - Q_p B f_p. \quad (\text{A.11})$$

Substitution into (A.8) gives $a + Bf_p = Q_p^{-1}X_p - Q_p^{-1}C_{Wf}$. Adding $B(\bar{f} - f_p)$ gives

$$\bar{h} = a + B\bar{f} = Q_p^{-1}X_p - Q_p^{-1}C_{Wf} + B(\bar{f} - f_p), \quad (\text{A.12})$$

which proves (16). The co-movement term appears because the intercept condition averages realized factor contributions through changing characteristic exposures. A simple average of the factor realizations need not preserve that weighting. For an approximate solution, define the L -vector of normal-equation error $u = \sum_t p_t \{X_t - W_t(a + Bf_{t+1})\}$. The exact right-hand side additionally contains $-Q_p^{-1}u$; the replication records the attained numerical error.

If W_t is constant across training dates, $C_{Wf} = 0$. If the factor mean uses the training weights, the final date-weight term also vanishes. In that case the complete coefficient equals pooled regression even when the loading matrix is held fixed. More generally, geometry can alter the fitted mean through its association with factor realizations. This supplies an estimation channel to test, rather than a guarantee that a dynamic ruler improves forecasts.

A.3 Forecast averaging and component retention

The following identities hold on any common evaluation observations. Let $W = Z'DZ \in \mathbb{R}^{L \times L}$, $X = Z'Dy \in \mathbb{R}^L$ and $Y_2 = y'Dy \in \mathbb{R}$ be their characteristic, cross-return and squared-return moments, with Z , D and y defined as above for this evaluation sample. Nonnegative stock weights make W positive semidefinite; no inverse is needed. The squared loss of a mean coefficient $h \in \mathbb{R}^L$ is $\ell(h) = Y_2 - 2X'h + h'Wh$. For fixed forecasts $h_A, h_B \in \mathbb{R}^L$, write $h_C = (h_A + h_B)/2$ and $d = (h_A - h_B)/2$. Expanding both losses gives

$$\ell(h_C + d) + \ell(h_C - d) = 2Y_2 - 4X'h_C + 2h_C'Wh_C + 2d'Wd \quad (\text{A.13})$$

$$= 2\ell(h_C) + 2d'Wd. \quad (\text{A.14})$$

Dividing by two and rearranging yields

$$\frac{\ell(h_A) + \ell(h_B)}{2} - \ell(h_C) = \frac{1}{4}(h_A - h_B)'W(h_A - h_B) \geq 0. \quad (\text{A.15})$$

Averaging reduces loss relative to the average loss of its two components by their squared signal disagreement. This algebraic gain requires no independence of model errors and does not guarantee that the average beats the better component.

For a projected coefficient $p \in \mathbb{R}^L$ and residual coefficient $r \in \mathbb{R}^L$, expand the same criterion at $p + r$:

$$\ell(p + r) = \ell(p) - 2X'r + 2p'Wr + r'Wr. \quad (\text{A.16})$$

Rearrangement gives

$$\ell(p) - \ell(p + r) = 2r'(X - Wp) - r'Wr, \quad (\text{A.17})$$

which proves the residual-benefit identity. The residual helps on these evaluation observations when its alignment with the projected forecast's error offsets the squared size of the added signal. Orthogonality under the historical formation Gram is not needed for this identity and does not force $p'Wr$ to vanish under a later evaluation Gram. The formula therefore measures the actual later benefit of using the residual signal, rather than inferring that benefit from its formation-period geometry.

B Objects and Computation

B.1 Local definitions and the monthly loss

This appendix gives the moment calculations underlying the common forecasting task. At formation month t , Z_t is the $N_t \times L$ ranked-characteristic matrix for the declared population, with $L = 153$ in the application. D_t is its diagonal $N_t \times N_t$ matrix of stock weights, normalized to sum to one. EW assigns equal stock weights; VW uses formation market capitalization. The N_t -vector y_{t+1} is the clipped next-month excess return. On the common observed-return population, define $W_t = Z_t' D_t Z_t$, $X_t = Z_t' D_t y_{t+1}$, and $Y_{2,t} = y_{t+1}' D_t y_{t+1}$, of dimensions $L \times L$, L and one. Characteristics and their second moments are dimensionless; y_{t+1} , X_t and a fixed L -vector of mean coefficients h have monthly return units. Holding the evaluation observations and weights fixed, the monthly squared loss expands as

$$(y_{t+1} - Z_t h)' D_t (y_{t+1} - Z_t h) = y_{t+1}' D_t y_{t+1} - 2h' Z_t' D_t y_{t+1} + h' Z_t' D_t Z_t h \quad (\text{B.1})$$

$$= Y_{2,t} - 2X_t' h + h' W_t h. \quad (\text{B.2})$$

This identity permits aggregate moments to reproduce the squared-loss evaluation without redistributing licensed stock rows. It also shows why every candidate must use the same moments: changing the outcome or stock weights changes the prediction task. Source validation of the moments remains a separate data requirement.

The formation-only Gram uses positive-capitalization formation stocks without reading or filtering on the next return. Ranks are formed on all stored formation observations, separately by characteristic, before the capitalization or return subset is imposed. Average ranks are divided by the count of nonmissing values plus one and centered at one half. A missing value, or a column with fewer than two nonmissing observations, receives zero. This preserves a common encoding when estimation and evaluation weights change.

B.2 Residual projection and an independent calculation

Let $B \in \mathbb{R}^{L \times K}$ be a full-rank loading matrix, $g \in \mathbb{R}^L$ a complete mean target and $Q \in \mathbb{R}^{L \times L}$ a positive-definite measurement Gram. Holding all three fixed, the residual is $r = g - B(B'QB)^{-1}B'Qg$. For an independent numerical calculation, let $C \in \mathbb{R}^{L \times L}$ satisfy $C'C = Q$ and let $U \in \mathbb{R}^{L \times K}$ be an orthonormal basis for the columns of CB , obtained by QR decomposition. Then

$$r'Qr = \|(I_L - UU')Cg\|_2^2. \quad (\text{B.3})$$

Multiplication by C converts the weighted projection to an ordinary Euclidean projection. The residual magnitude can therefore be checked without reusing the inverse formula. The replication calculations also check $B'Qr = 0$ and invariance to factor-space rotations and intercept transfers. These are algebra and numerical checks, not sampling-distribution results.

B.3 Why quadratic penalties must move with a representation

Let $A \in \mathbb{R}^{L \times L}$ be nonsingular and set $\tilde{Z}_t = Z_t A$. For a coefficient $b \in \mathbb{R}^L$, the same linear signal uses $\tilde{b} = A^{-1}b$. If an estimation criterion consists of a fitted-signal loss and a quadratic penalty with symmetric positive-semidefinite $H \in \mathbb{R}^{L \times L}$, substitution gives

$$Z_t b = \tilde{Z}_t \tilde{b}, \quad b' H b = \tilde{b}' A' H A \tilde{b}. \quad (\text{B.4})$$

The transformed optimization problem has the same feasible signals and objective values if the penalty matrix becomes $A' H A$ and any other constraints are transported too. The matrix H here includes the penalty's scalar multiplier; a data-dependent normalization of that multiplier must be transported consistently. Minimizers then correspond under the invertible map, whether or not the coefficient minimizer is unique. Setting the penalty to the identity in both representations is an additional estimator choice and generally breaks this correspondence. For a fixed signal, in contrast, transporting the Gram to $A' Q A$ always preserves its measured norm.

For intuition, measure one characteristic in units one hundred times larger. The coefficient needed for the same signal becomes one hundred times smaller. An unchanged Euclidean penalty now penalizes that signal less. The information in the characteristic has not changed; the estimator's relative shrinkage has. This is why the paper fixes the rank encoding for its baseline ridge comparisons. Appendix H then measures the forecasting effect of deliberately changing the regularization rule.

B.4 The crossing decomposition

The crossing comparison replaces the mean and the space one at a time. Its symmetric accounting averages the two possible replacement orders, so the result does not privilege one path through the table. For a fixed Q , define the scalar squared residual $\psi_{s,m} = \rho_Q(B_s, \bar{h}_m)^2$ in basis points squared, where s indexes the EW or VW loading space and m indexes the EW or VW fitted mean. Let $\Delta = \psi_{\text{EW},\text{EW}} - \psi_{\text{VW},\text{VW}}$ be the diagonal gap. The symmetric space contribution A_B and mean contribution A_h , both in basis points squared, are

$$A_B = \frac{1}{2}[(\psi_{\text{EW},\text{EW}} - \psi_{\text{VW},\text{EW}}) + (\psi_{\text{EW},\text{VW}} - \psi_{\text{VW},\text{VW}})], \quad (\text{B.5})$$

$$A_h = \frac{1}{2}[(\psi_{\text{EW},\text{EW}} - \psi_{\text{EW},\text{VW}}) + (\psi_{\text{VW},\text{EW}} - \psi_{\text{VW},\text{VW}})]. \quad (\text{B.6})$$

Expanding and cancelling the off-diagonal terms gives $A_B + A_h = \Delta$. Reported shares divide each contribution by Δ when it is nonzero. This averages the two paths through the crossing. A contribution can be negative, and the other can exceed the diagonal gap: the mean and space changes can work in opposite directions. These are accounting shares of the fitted-object contrast, not causal shares of the true economic risk premium.

B.5 Ridge controls and the monthly inverse domain

The direct mean controls test whether the factor construction improves on simpler estimates of the same forecasting target. For a training window, let $\bar{W} \in \mathbb{R}^{L \times L}$ and $\bar{X} \in \mathbb{R}^L$ be its pooled characteristic and cross-return moments, and let I_L be the identity matrix. Pooled ridge solves $(\bar{W} + \lambda c I_L)g = \bar{X}$ for the mean coefficient $g \in \mathbb{R}^L$, where $\bar{W}$ and $\bar{X}$ average training moments with weights proportional to the common stock counts, $c = \text{tr}(\bar{W})/L$, and $\lambda \in \{0, 0.01, 0.1, 1, 10\}$ is dimensionless. The last 60 training months form the validation block. Each candidate is fitted on the earlier training months and evaluated with equal-month validation loss; the selected value is refitted using the complete training window.

The monthly ridge control first computes the L -vector $u_{t+1}(\gamma) = (W_t + \gamma c_t I_L)^{-1} X_t$, where $c_t = \text{tr}(W_t)/L$ and $\gamma \in \{0.01, 0.1, 1, 10\}$ is a dimensionless penalty parameter. For each candidate, the equal-date mean of pre-validation monthly coefficients predicts the last 60 training returns. The selected γ is then applied to all training months, whose coefficient mean becomes the forecast. Unregularized monthly least squares is eligible only if every training month's smallest Gram eigenvalue exceeds 10^{-10} times its largest. This criterion fails in 99 early months of the full sample. With $c_t > 0$, positive ridge penalties keep these monthly regressions well defined, but also change the estimator. Monthly ridge is therefore reported as a separate benchmark; it is not relabeled as the unavailable unregularized

monthly regression.

C Decision Controls and Complete Sensitivities

The model-selection results depend on the forecast information set, the treatment of near ties and the alternatives available to the user. This appendix records those conventions and changes one design choice at a time while retaining the common forecasts and evaluation outcomes.

C.1 Timing, ties and ranking statistics

The annual origin label is the last training formation month, February of each year 2017–2025. March training returns are available before the first April target return. The primary ruler uses the 60 formation months ending at that label. Thus it uses a slightly lagged but available formation window. Target means and factor estimates use only completed training outcomes. Recomputing a diagnostic from data truncated at its origin, and then perturbing later outcomes, checks that the diagnostic is determined by information available at that origin.

An RMS difference of at most 10^{-6} bp is treated as a numerical tie; the dimensionless own-residual share uses tolerance 10^{-10} . Ties select a 50/50 forecast average. First-origin full/fixed-space comparisons are structural ties because their feasible spaces coincide. They are omitted from ranking accuracy rather than counted as successes. Balanced accuracy is the mean of the conditional correct-selection rates for the two realized winner classes and is undefined when a class is absent. Spearman correlations are descriptive correlations of signed diagnostic and subsequent loss differences within a sequence, without reported significance tests.

C.2 Recorded choices and accumulated loss

Table C.1 records the number of choices, realized winners and correct rankings behind the selection comparison. Figure C.1 displays the same annual decisions so that the constant preference from using each model’s own mean can be compared with changing subsequent winners. Together they distinguish a successful fixed candidate from a diagnostic that contributes new selection information.

The magnitude of a wrong choice matters as well as its frequency. Figure C.2 adds each selector’s monthly loss relative to the equal average of the two complete forecasts. An upward movement means that the selector loses ground to the average during those months.

EW evaluation									
Own mean	VW	VW	VW	VW	VW	VW	VW	VW	VW
Common ridge	EW	EW	EW	EW	EW	EW	VW	VW	EW
Common pooled	VW	VW	VW	VW	VW	VW	VW	VW	VW
Future winner	VW	VW	VW	VW	EW	EW	VW	EW	EW
VW evaluation									
Own mean	VW	VW	VW	VW	VW	VW	VW	VW	VW
Common ridge	VW	VW	VW	EW	VW	VW	VW	VW	VW
Common pooled	VW	VW	VW	VW	VW	VW	VW	VW	VW
Future winner	VW	VW	EW	VW	EW	EW	VW	VW	VW
	2017	2018	2019	2020	2021	2022	2023	2024	2025
Annual decision origin									

Figure C.1: Diagnostic choices and subsequent winners.

Rows compare diagnostic choices with the subsequent lower-loss candidate at nine annual origins. Each choice is between complete three-factor forecasts trained using equal-stock (EW) or formation-capitalization (VW) weights, with expanding history. The own-mean diagnostic projects each model's own fitted mean and always selects VW. The common-target diagnostic projects the same historical ridge mean onto both spaces. Both use a common preceding-60-month formation Gram under the panel's evaluation weights. The winner is the candidate with smaller next-block squared loss. Target months are April 2017–December 2025, with nine months in the last block and twelve in the others. The display distinguishes a constant model preference from origin-specific information about future performance.

Table C.1: Diagnostic choices and subsequent winners

Diagnostic	EW / VW choices	Correct	Balanced (%)	Switches
<i>Equal-stock evaluation (EW)</i>				
Model's own mean	0/9	5/9	50.0	0
Own-mean residual share	0/9	5/9	50.0	0
Common ridge mean	7/2	4/9	47.5	2
Common pooled mean	0/9	5/9	50.0	0
<i>Capitalization-weighted evaluation (VW)</i>				
Model's own mean	0/9	6/9	50.0	0
Own-mean residual share	0/9	6/9	50.0	0
Common ridge mean	1/8	5/9	41.7	2
Common pooled mean	0/9	6/9	50.0	0

Each row contains nine annual decisions, 2017–2025, between complete EW-trained and VW-trained three-factor forecasts with expanding training. The choice column gives the number of times each candidate is selected; Correct counts selections with lower subsequent squared loss; Switches counts changes between successive choices. Balanced accuracy averages the correct-selection rates within the two realized winner classes, giving a constant selector 50% when both classes occur. A smaller diagnostic selects the next block's complete forecast. Own-mean diagnostics use each model's fitted mean; common diagnostics project the same historical ridge or unpenalized pooled mean onto both spaces. Both candidates share the preceding-60-month formation Gram under the stated evaluation weights. The 105 target months run from April 2017 to December 2025: eight twelve-month blocks and a final nine-month block. These are descriptive counts on the shared history.

The curves show when the final disadvantage accumulates; their common positive endpoints agree with the aggregate comparison in Table 3.

C.3 Reference targets and overlap

A common reference mean fixes the object compared across spaces, but its estimation history and the measurement window can still affect the ordering. Table C.2 reports every EW/VW factor-count and window combination under the primary common ridge target. Table C.3 changes only the Gram window. A further target uses the preceding origin's ridge mean. It is available from 2018 and shares old observations with the current fit. It is therefore a lagged historical reference, not an independent or cross-fitted mean estimate. The unpenalized pooled target and own-residual share are also retained in the result files.

The complete-year sensitivity uses April 2017–March 2025, omitting the final nine-month block. Comparisons involving a preceding forecast record use April 2018–December 2025, a common 93-month overlap. Leave-one-origin-out summaries omit each annual block in turn and recompute the pooled loss ratio on the retained months. These operations describe con-

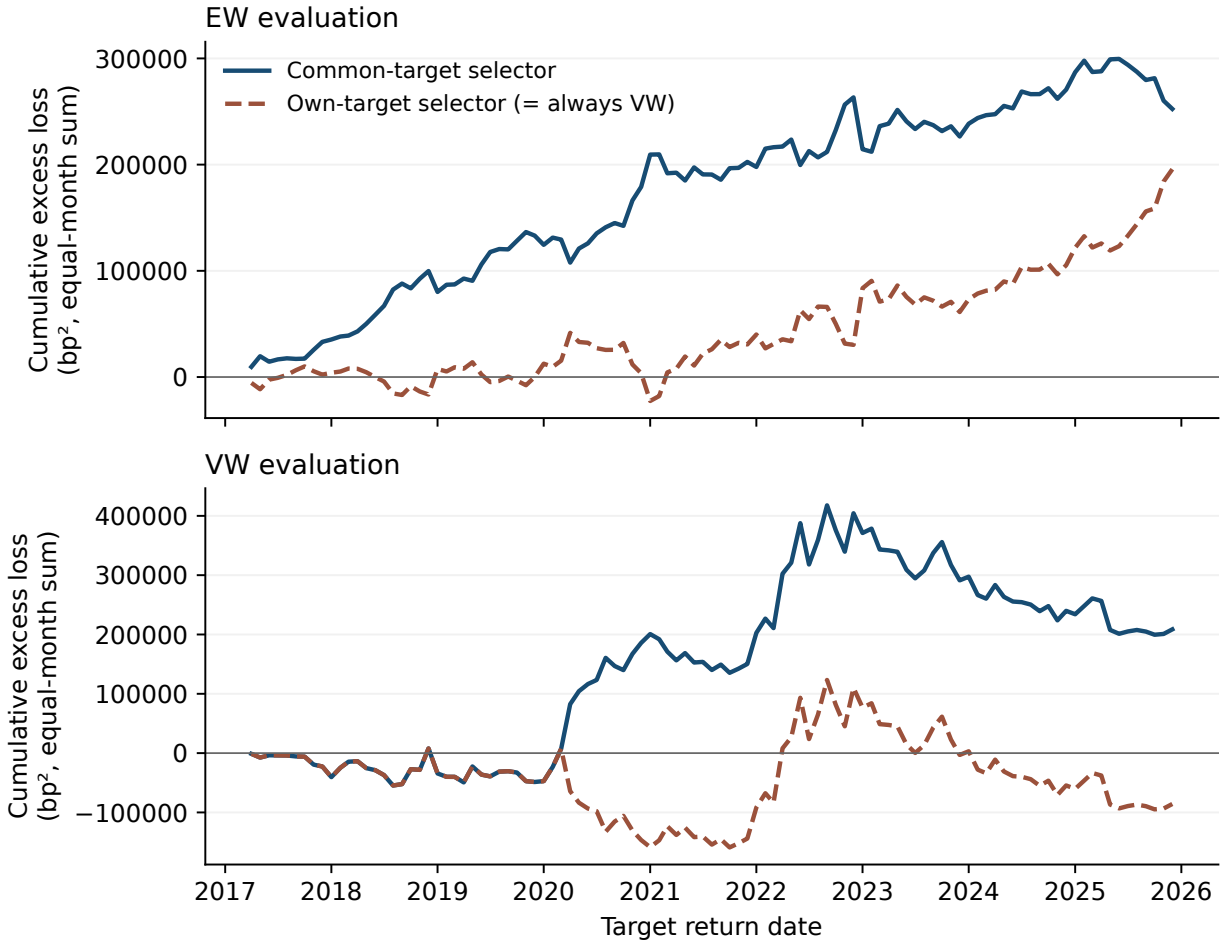


Figure C.2: Cumulative loss relative to a simple forecast average.

Each curve accumulates a selector's monthly weighted squared loss minus the loss of equally averaging the complete EW/VW factor forecasts. Positive values mean that the selector has accumulated more forecast error than averaging. Units are basis points squared, summed with equal month weights. The three-factor expanding design uses identical target observations from April 2017 to December 2025. EW evaluation weights stocks equally and VW uses formation capitalization. Selection based on each model's own mean always chooses the VW-trained forecast; common-target selection compares both spaces against the same historical ridge mean. The paths locate when relative forecast losses accrue and are not cumulative investment returns.

Table C.2: Common-target selection across factor counts and windows

Window	K	Eval.	Correct	Selector	Average	Ridge
<i>Last three columns: prediction R^2 (%)</i>						
Expanding	1	EW	1/9	0.3200	0.4411	0.3972
Expanding	1	VW	2/9	0.6183	0.9850	0.8550
Expanding	3	EW	4/9	0.3612	0.4396	0.3972
Expanding	3	VW	5/9	0.7326	0.9569	0.8550
Expanding	5	EW	5/9	0.3154	0.4104	0.3972
Expanding	5	VW	6/9	0.8855	0.9049	0.8550
Rolling 240	1	EW	8/9	0.6100	0.5457	0.6353
Rolling 240	1	VW	5/9	1.0741	1.0732	0.7648
Rolling 240	3	EW	8/9	0.6125	0.5374	0.6353
Rolling 240	3	VW	6/9	1.1550	1.1479	0.7648
Rolling 240	5	EW	8/9	0.6164	0.5259	0.6353
Rolling 240	5	VW	5/9	1.1326	1.1479	0.7648

The common-target selector compares the two factor spaces against one historical pooled ridge mean, using the preceding-60-month formation Gram. Correct counts selections with lower subsequent squared loss among nine annual decisions. Average equally combines the complete EW- and VW-trained factor forecasts; Ridge directly forecasts using the evaluation weights. EW evaluation gives stocks equal weight and VW uses formation capitalization. All prediction- R^2 levels use the common 105 target months, April 2017–December 2025. Rolling windows contain 240 training months. The diagnostic based on each model's own mean selects VW at all nine origins in all twelve sequences. These sequences vary the specification on the same target history.

Table C.3: Sensitivity to the measurement window

Evaluation	Gram window	Correct	Balanced (%)	R^2 (%)
EW	Last 60 months	4/9	47.5	0.3612
EW	Training window	6/9	70.0	0.3847
VW	Last 60 months	5/9	41.7	0.7326
VW	Training window	5/9	41.7	0.7326

Three-factor expanding models are selected using a common historical ridge mean. Only the averaging window for the formation Gram changes; the candidate forecasts, 105 target months and evaluation losses are held fixed. Both windows use information available at the annual origin. EW and VW denote equal-stock and formation-capitalization evaluation. Correct counts the lower-loss candidate selected in nine decisions; Balanced is accuracy averaged within the two realized winner classes. The final column is the selected forecast's prediction- R^2 level. Comparing the two rows within each evaluation shows whether the choice is sensitive to the history used to measure residuals.

centration in the observed history; they do not provide confidence coverage under sequential estimation.

C.4 Past performance and amplitude controls

A past-performance rule chooses the candidate label with lower loss in the preceding completed annual forecast block, then issues that label’s current forecast. It uses earlier issued predictions rather than fitting the choice against the next block. A separate amplitude control asks whether the original diagnostic anticipates rankings after a simple available calibration.

For candidate j , let $h_{j,o-1} \in \mathbb{R}^L$ be its coefficient issued at the previous annual origin, and let $\mathcal{H}_{o-1}$ contain its completed target months. The scalar calibration is

$$c_{j,o} = \text{clip}_{[0,1]} \left\{ \frac{\sum_{t+1 \in \mathcal{H}_{o-1}} X'_t h_{j,o-1}}{\sum_{t+1 \in \mathcal{H}_{o-1}} h'_{j,o-1} W_t h_{j,o-1}} \right\}. \quad (\text{C.1})$$

The operator truncates its argument to the interval $[0, 1]$. The expression minimizes the previous block’s squared loss over that interval. Both sums use the current declared evaluation weights. Apply the resulting coefficient to the current mean $h_{j,o}$ and leave the original diagnostic fixed. The first origin uses one; the common-overlap comparison excludes that initialization. This control asks whether the original diagnostic still ranks the models usefully after both complete means receive an ordinary amplitude adjustment. The twelve-month record can make the fitted scalars noisy, so a changed ordering identifies the consequence of this calibration rule, rather than a general mechanism of geometry.

C.5 The updating grid

Updating a latent factor space adds estimation work and can change both the decomposition and the complete mean. The fixed-space control asks whether that update improves subsequent prediction when the intercept and factor history can still adapt. Tables C.4 and C.5 give the full 24-cell forecast grid behind this comparison. The fixed space is the loading space estimated at the first origin for that factor count, training window and estimation weighting. Its intercept and factor history are re-estimated thereafter. Unrestricted intercepts remain L -dimensional in both models. Additional machine-readable results retain the frozen first-origin forecast, stock-count-weighted factor means, calendar subperiods and eligible unregularized monthly controls.

An infeasible annual component-choice rule is also recorded: it chooses retrospectively the better of the two complete component forecasts in each annual block. This rule is an

Table C.4: Expanding forecast comparisons

K	Train	Eval.	Full	Fixed	Pooled	Monthly	Factor+ridge
$Prediction$	R^2	(%)					
1	EW	EW	0.3200	0.3188	0.3972	0.3112	0.4229
1	EW	VW	0.6183	0.6194	0.8181	0.7126	0.8473
1	VW	EW	0.4039	0.4107	0.3582	0.2910	0.4138
1	VW	VW	1.0035	0.9935	0.8550	1.0435	0.9928
3	EW	EW	0.3605	0.3732	0.3972	0.3112	0.4361
3	EW	VW	0.5678	0.6127	0.8181	0.7126	0.8015
3	VW	EW	0.3788	0.3835	0.3582	0.2910	0.3957
3	VW	VW	1.0491	1.0591	0.8550	1.0435	1.0099
5	EW	EW	0.3724	0.3581	0.3972	0.3112	0.4425
5	EW	VW	0.6019	0.6060	0.8181	0.7126	0.8167
5	VW	EW	0.3152	0.3146	0.3582	0.2910	0.3644
5	VW	VW	0.9472	1.0122	0.8550	1.0435	0.9611

Entries are prediction- R^2 levels in percent on 105 target months, April 2017–December 2025; larger values indicate smaller forecast losses. Train specifies estimation weights and Eval. specifies loss weights: EW gives stocks equal weight, while VW uses formation capitalization. Full re-estimates the loading space at each origin; Fixed retains the first-origin space while updating the intercept and factor history. Pooled is pooled ridge; Monthly averages the monthly ridge coefficients. Factor+ridge is the equal average of Full and Pooled under the same training weights, distinct from the EW/VW factor average elsewhere. All tuning uses historical observations, and factor means use equal date weights.

upper benchmark for choosing one of those two components with that block structure. It is not an upper benchmark for all forecasts. In particular, an average may have smaller squared loss than either component, so it can outperform that retrospective component choice.

Table C.5: Rolling 240-month forecast comparisons

K	Train	Eval.	Full	Fixed	Pooled	Monthly	Factor+ridge
<i>Prediction R^2 (%)</i>							
1	EW	EW	0.6100	0.5883	0.6353	0.5242	0.6736
1	EW	VW	0.9693	0.9678	0.9969	1.1100	1.0848
1	VW	EW	0.2842	0.3373	0.3282	0.3408	0.3643
1	VW	VW	0.8216	0.8237	0.7648	1.1504	0.8912
3	EW	EW	0.6125	0.6203	0.6353	0.5242	0.6711
3	EW	VW	1.0650	1.0637	0.9969	1.1100	1.1205
3	VW	EW	0.3205	0.3455	0.3282	0.3408	0.3686
3	VW	VW	0.9898	0.9669	0.7648	1.1504	0.9628
5	EW	EW	0.6164	0.6270	0.6353	0.5242	0.6728
5	EW	VW	1.1055	1.1186	0.9969	1.1100	1.1371
5	VW	EW	0.3021	0.3261	0.3282	0.3408	0.3549
5	VW	VW	0.9731	1.0205	0.7648	1.1504	0.9494

Entries are prediction- R^2 levels in percent on 105 target months, April 2017–December 2025; larger values indicate smaller forecast losses. Train specifies estimation weights and Eval. specifies loss weights: EW gives stocks equal weight, while VW uses formation capitalization. Full re-estimates the loading space at each origin; Fixed retains the first-origin space while updating the intercept and factor history. Pooled is pooled ridge; Monthly averages the monthly ridge coefficients. Factor+ridge is the equal average of Full and Pooled under the same training weights, distinct from the EW/VW factor average elsewhere. All tuning uses historical observations, and factor means use equal date weights.

D A Controlled Geometry-by-Mean Experiment

The empirical identities allow characteristic geometry and factor means to move together, but that algebra does not establish a source of predictability. This simulation varies those two ingredients separately: whether characteristic second moments change, and whether an observed state predicts the factor mean. Keeping the loading known isolates mean estimation from latent-space estimation. There are $L = 8$ characteristics, $K = 1$ known factor loading, 240 training and 120 prediction months, and an artificial cross-sectional size of 1,000. Let $e_j \in \mathbb{R}^8$ denote coordinate vector j , set $B = e_1$ and $a = 0.004e_2$, and draw an independent equiprobable state $s_t \in \{-1, 1\}$ for each month. The factor is $f_{t+1} = 0.004 + \delta s_t + \varepsilon_{t+1}$, where ε_{t+1} is independent normal noise with standard deviation 0.04 and δ equals zero or 0.008.

Let $D = \text{diag}(0.14, 0.025, 0.07, 0.06, 0.05, 0.045, 0.035, 0.03)$. The 8×8 rotation matrix R_s rotates the first two coordinates by $s\pi/6$ and leaves the other coordinates unchanged. Set $Q_s = R_s D R_s'$ and $\bar{Q} = (Q_{-1} + Q_1)/2$. The geometry is either $W_t = Q_{s_t}$ or the constant

$W_t = \bar{Q}$. The simulated return moment is

$$X_t = W_t(a + Bf_{t+1}) + \eta_t, \quad \eta_t \mid W_t \sim N(0, 0.1^2 W_t/1000), \quad (\text{D.1})$$

independently across months and independently of factor shocks. These equations specify the moments needed for estimation directly: W_t measures characteristic exposures and X_t their return covariation, with sampling noise $\eta_t \in \mathbb{R}^8$. The state is observed at prediction time; no empirical state classification or latent loading estimation is hidden in this experiment.

Each replication compares the complete mean from a fixed known loading fit, pooled regression and a mean conditioned on the state. The conditional procedure averages fitted factors separately in the two training states and combines the relevant average with the estimated intercept. The test target is the known conditional mean $a + B(0.004 + \delta s_t)$. Errors use the test-state Gram and are reported in basis points squared. Common random draws across the four cells isolate changes in geometry and the factor mean. The seed is 2026092107, with 500 replications per cell.

Table D.1 shows that state conditioning adds noise when the factor mean is constant, including when geometry rotates. When the mean varies with the observed state, conditioning improves on the unconditional means even with constant geometry. The useful information is the observed state's connection to the conditional mean: rotation alone is neither necessary nor sufficient for the conditional forecast to improve in this design. Economically, a changing exposure distribution does not by itself imply a changing expected factor payoff that a forecast can exploit. Because loadings and the state are known here, the simulation illustrates this mechanism without identifying its empirical importance in the U.S. data.

Table D.1: Geometry and conditional means: a controlled simulation

Geometry	Factor mean	Fixed space	Pooled	State mean
<i>Conditional-mean signal error (bp squared)</i>				
Constant	Constant	109.77	109.77	193.86
Constant	State dependent	826.95	826.95	193.86
Rotating	Constant	112.74	141.67	193.86
Rotating	State dependent	830.46	560.77	193.86

Entries are mean squared errors in forecasting the known conditional-mean signal, in basis points squared, averaged over 500 replications per cell; lower is better. Each simulation has eight characteristics, one known factor loading, 240 training and 120 prediction months, and 1,000 stocks per month. An observed two-state variable either rotates the first two characteristic directions by plus/minus 30 degrees or leaves the Gram fixed; independently, the factor mean is constant or depends on that state. Fixed space estimates the factor mean in the known loading space; Pooled estimates the characteristic mean coefficient directly; State mean conditions the factor-mean estimate on the observed state. The crossed design distinguishes the effect of changing characteristic geometry from the benefit and estimation cost of conditioning on a state. The outcome is simulated signal error, rather than empirical realized-return loss.

E Population Accounting and Representation Controls

The population checks distinguish two questions: where a fixed residual signal is concentrated, and which stocks contribute to a forecast’s subsequent performance. The representation check then holds the attainable signals fixed and asks whether a coordinate change alters the estimator through its penalty.

E.1 Group moments and the forecast population

Groups are the stored formation-size labels mega, large, small, micro, nano and unclassified. The exercise does not reconstruct their breakpoints. The underlying security vintage, including its size labels, is inherited from the source panel. Unclassified has zero mass in these calculations. The larger-stock reporting group combines mega, large and small. This is a transparent aggregation of stored labels, rather than a new investability screen.

For a group c , let $\mathcal{I}_{c,t}$ be its set of included stocks at month t and $d_{i,t}$ the stock’s weight

in the full evaluation population. Define

$$W_{c,t} = \sum_{i \in \mathcal{I}_{c,t}} d_{i,t} z_{i,t} z'_{i,t} \in \mathbb{R}^{L \times L}, \quad (\text{E.1})$$

$$X_{c,t} = \sum_{i \in \mathcal{I}_{c,t}} d_{i,t} z_{i,t} y_{i,t+1} \in \mathbb{R}^L, \quad (\text{E.2})$$

$$Y_{2,c,t} = \sum_{i \in \mathcal{I}_{c,t}} d_{i,t} y_{i,t+1}^2, \quad m_{c,t} = \sum_{i \in \mathcal{I}_{c,t}} d_{i,t}. \quad (\text{E.3})$$

The last two objects are scalars. The group weight mass is $m_{c,t}$; it is not yet normalized to one. Summing any of these group moments over the partition recovers its full-population counterpart. For an issued coefficient $h \in \mathbb{R}^L$, the unnormalized group loss is $L_{c,t}(h) = Y_{2,c,t} - 2X'_{c,t}h + h'W_{c,t}h$. It is additive across groups, which makes it appropriate for accounting for an aggregate model-performance difference.

For a user forecasting a particular group, the relevant question is performance conditional on belonging to that group. Table E.2 therefore describes performance within a group. Divide both $L_{c,t}(h)$ and $Y_{2,c,t}$ by $m_{c,t}$ before summing over months and forming their ratio. This treats each month's conditional population equally. Table E.1 uses the unnormalized losses and the common aggregate denominator. These two normalizations answer the two practical questions separately: how accurately a model forecasts a particular group, and how much that group contributes to the advantage observed for the full population. Ranks and return-clipping thresholds remain defined using the full populations described in the main text.

Table E.1: Additive sources of aggregate forecast improvements

Group	Equal-stock evaluation		Cap-weight evaluation	
	VW-trained	Average	VW-trained	Average
<i>Contributions to R^2 gain over the EW-trained forecast (pp)</i>				
Larger stocks	0.06755	0.06028	0.47949	0.38603
Micro	-0.02761	0.01243	0.00171	0.00289
Nano	-0.02160	0.00644	0.00009	0.00020

Each entry is a stock group's contribution, in percentage points (pp), to the prediction- R^2 gain over the complete EW-trained forecast. Positive values favor the column's alternative. Average equally combines the complete EW/VW forecasts. Larger stocks combine mega, large and small; micro and nano remain separate. Group loss differences retain full-population weights over the same 105 target months and are divided by the full-population zero-forecast loss. Contributions therefore sum to the aggregate gain before rounding. This additive accounting differs from the within-group renormalization in Table E.2.

Table E.2: Group weights and diagnostic-selector outcomes

Group	Weight (%)	EW-trained	VW-trained	Selector
<i>Last three columns: within-group prediction R^2 (%)</i>				
<i>Equal-stock evaluation</i>				
Larger stocks	50.640	-0.0698	0.1681	-0.0188
Micro	35.676	0.1289	0.0725	0.1270
Nano	13.684	0.9010	0.7773	0.8719
<i>Capitalization-weighted evaluation</i>				
Larger stocks	99.021	0.5850	1.0809	0.7534
Micro	0.943	0.0092	0.0354	0.0573
Nano	0.036	0.8396	0.6364	0.7766

Group weights are equal-month mean shares of the full evaluation population over 105 target months, April 2017–December 2025. Larger stocks combine mega, large and small; micro and nano remain separate. Forecast columns evaluate the three-factor expanding EW-trained or VW-trained model, using equal-stock or formation-capitalization training respectively. The selector compares both spaces against the same historical ridge mean. It makes one whole-population choice at each annual origin, carried unchanged into every group. Within-group R^2 renormalizes stock weights separately in each month and group, so these levels need not aggregate to the full-population result.

The six-group monthly moments were rebuilt from the licensed panel in the earlier data-construction audit, for formation March 2017–November 2025. Their sums reconcile to the common-population moments to 10^{-12} in absolute units. For three separated months, direct stock-level losses under a fixed nontrivial coefficient vector agree with the moment formula to less than 10^{-16} . The replication workflow uses aggregate moments and their manifest. Rebuilding those moments requires access to the licensed stock panel; raw stock-level records are not redistributed with the manuscript. These checks validate the aggregation, not the unresolved return-vintage and delisting issues described below.

Table E.3 provides the weights and residual shares behind the main concentration figure. Each full-sample residual is held fixed while its Gram norm is partitioned. The forecast-period group masses in Table E.2 differ because the dates differ. Table E.4 repeats the mean-space crossing at each annual origin using only the historical estimates and the common recent formation Gram.

E.2 Recoding, validation and equivalent penalties

Re-expressing the same characteristics cannot add return information, but it can change a ridge forecast if the penalty is specified anew in the new coordinates. To identify that channel, fix the data and attainable linear signals and compare a new identity penalty with

Table E.3: Population weight and residual concentration

Group	Equal-stock measurement		Cap-weight measurement	
	Stock weight	Residual share	Stock weight	Residual share
	<i>Shares (%) for the EW-trained model</i>			
Mega	7.59	4.77	74.01	68.54
Large	13.57	9.58	16.34	17.52
Small	20.79	17.43	6.65	8.82
Micro	39.45	43.68	2.74	4.63
Nano	18.60	24.54	0.26	0.49

This table compares each size group's population weight with its share of the full-sample EW-trained model's squared residual norm. Each measurement uses one full-population projection, then attributes its fixed residual to the groups. Each column sums to 100% before rounding. Formation dates are July 1963–November 2025, and no observations are unclassified. A residual share above the group's stock-weight share indicates a disproportionate contribution to the fitted residual. These are additive full-population shares, not separately estimated within-group residual RMS values.

Table E.4: The mean-space crossing at each historical origin

Origin	Own fitted-mean RMS (bp)		Mean-change share (%)	
	EW-trained	VW-trained	EW measurement	VW measurement
2017	120.60	83.01	100.71	100.76
2018	118.92	81.96	100.89	100.84
2019	117.53	81.03	100.51	100.14
2020	116.63	80.14	98.84	97.37
2021	116.45	79.04	98.52	96.40
2022	115.22	79.46	99.14	97.58
2023	116.77	78.74	97.33	96.27
2024	118.25	77.84	96.80	95.46
2025	119.38	76.45	97.54	96.76

The first two columns measure each model's own fitted-mean residual under a common EW formation Gram, in monthly basis points (bp). The last two give the mean change's symmetric two-path attribution of the squared diagonal gap, averaging the contributions obtained by changing the mean before and after the space. EW and VW measurement use equal-stock and formation-capitalization Grams, respectively, averaged over the preceding 60 months. A share above 100% means that the space contribution partly offsets the mean contribution. Every mean, space and measurement matrix is available at the named origin. Repeating the crossing over these origins checks whether the full-sample descriptive pattern persists across historical information sets.

the original penalty transported through the coordinate map. Let $Q \in \mathbb{R}^{L \times L}$ and $x \in \mathbb{R}^L$ be the pooled training moments under normalized stock-count date weights. For a nonsingular transformation $A \in \mathbb{R}^{L \times L}$, define $Q_A = A'QA$, $x_A = A'x$ and $c_A = \text{tr}(Q_A)/L$. Identity ridge in the new coordinates returns, in the original coordinates,

$$h_A = A(Q_A + \lambda c_A I_L)^{-1} x_A = [Q + \lambda c_A A^{-T} A^{-1}]^{-1} x. \quad (\text{E.4})$$

The equality follows by premultiplying and postmultiplying the normal equation by the inverse coordinate map. It shows the implied original-coordinate penalty explicitly. In contrast, if $c_0 = \text{tr}(Q)/L$ and the original identity penalty is transported, the corresponding equation is

$$h_{\text{transported}} = A[Q_A + \lambda c_0 A' A]^{-1} x_A = [Q + \lambda c_0 I_L]^{-1} x. \quad (\text{E.5})$$

The transport check fixes the baseline selected λ and compares these two computations numerically at every origin. All five recodings reproduce the baseline to less than 4×10^{-12} bp in monthly stock-weighted signal RMS.

The nontrivial maps are $A = \text{diag}(Q_{11}^{-1/2}, \dots, Q_{LL}^{-1/2})$ and the symmetric $A = Q^{-1/2}$. Positive definiteness is checked before either map is applied. Common scaling uses $A = 10I_L$; the orthogonal control uses a fixed QR rotation from Gaussian entries with seed 20260922. The last 60 training returns form the inner validation block. Fit the data-dependent transformation on the earlier inner block, compare the five ridge parameters, then refit both transformation and coefficients on all outer training observations. Future observations enter only evaluation.

For whitening, $Q_A = I_L$ and $c_A = 1$, so $h_A = (1 + \lambda)^{-1} Q^{-1} x$. Whitening followed by identity ridge therefore scales the pooled coefficient uniformly. Its apparent geometric complexity has a simple forecasting consequence: it changes the amplitude of the unpenalized pooled signal. The standardization and whitening comparisons use different penalties, whereas their transported versions test the exact same estimator in different coordinates.

F Residual Retention and Available Calibration

F.1 Loss accounting

The calibration experiment needs a criterion for deciding whether retaining a residual improves the issued forecast. The derivation follows by holding the two components fixed and expanding the same subsequent squared loss used to judge every rule. Fix an issued projected coefficient $p \in \mathbb{R}^L$ and residual coefficient $r \in \mathbb{R}^L$, with complete forecast $p + r$. For

a scalar $\eta \in [0, 1]$, expansion of the quadratic loss gives

$$\ell(p + \eta r) = Y_2 - 2X'(p + \eta r) + (p + \eta r)'W(p + \eta r) \quad (\text{F.1})$$

$$= \ell(p) - 2\eta r'(X - Wp) + \eta^2 r'Wr. \quad (\text{F.2})$$

Here W is the evaluation Gram, X the characteristic–return moment and Y_2 the zero-forecast loss for that month. The cross term uses the symmetry of W . Setting $\eta = 1$ yields the main-text benefit identity: the residual helps when it aligns sufficiently with the projected forecast’s errors to compensate for its squared signal size. Table F.1 adds its terms over the common forecast horizon and divides by the total zero-forecast loss; multiplication by 100 puts them in prediction- R^2 percentage points. All four monthly identities are independently reconciled before aggregation.

Table F.1: Why retaining the full residual can help or hurt

Training / evaluation	Twice alignment	Size cost	Net benefit	Omit-one range
<i>Contributions to prediction R^2 (percentage points)</i>				
EW / EW	0.6215	0.4696	0.1518	[0.0816, 0.2643]
EW / VW	0.8831	0.8594	0.0237	[-0.1115, 0.1238]
VW / EW	0.3298	0.2046	0.1252	[0.0746, 0.1612]
VW / VW	0.3435	0.3823	-0.0388	[-0.1049, 0.1079]

The table decomposes the prediction gain from retaining the complete residual rather than using only the projected forecast, as in equation (15). Twice alignment is the residual's contribution to correcting the projected forecast's errors; Size cost is its squared signal magnitude. Net benefit is the first contribution minus the second, up to rounding. All terms share the total zero-forecast loss denominator over 105 target months, April 2017–December 2025. EW and VW are equal-stock and formation-capitalization weights, respectively, with training listed before evaluation. Projections use the corresponding preceding-60-month formation Gram. The last column recomputes net benefit after omitting each annual evaluation block in turn; the resulting sensitivity range is not a confidence interval.

The fixed grid contains 0, 1/4, 1/2, 3/4, 1 for every training weight, evaluation weight and ruler. The identity-matrix comparator uses the original formation rank coordinates. It is not relabeled as an invariant geometric object. A third control applies each fraction to the entire complete mean. At fraction one the three forecasts coincide; at zero, only whole-mean shrinkage issues the zero forecast. Lines in Figure F.1 connect five evaluated rules; they summarize the observed loss trade-off without estimating a continuous optimum or an uncertainty band. The machine-readable summary also reports the first 96 target months, excluding the incomplete final block.

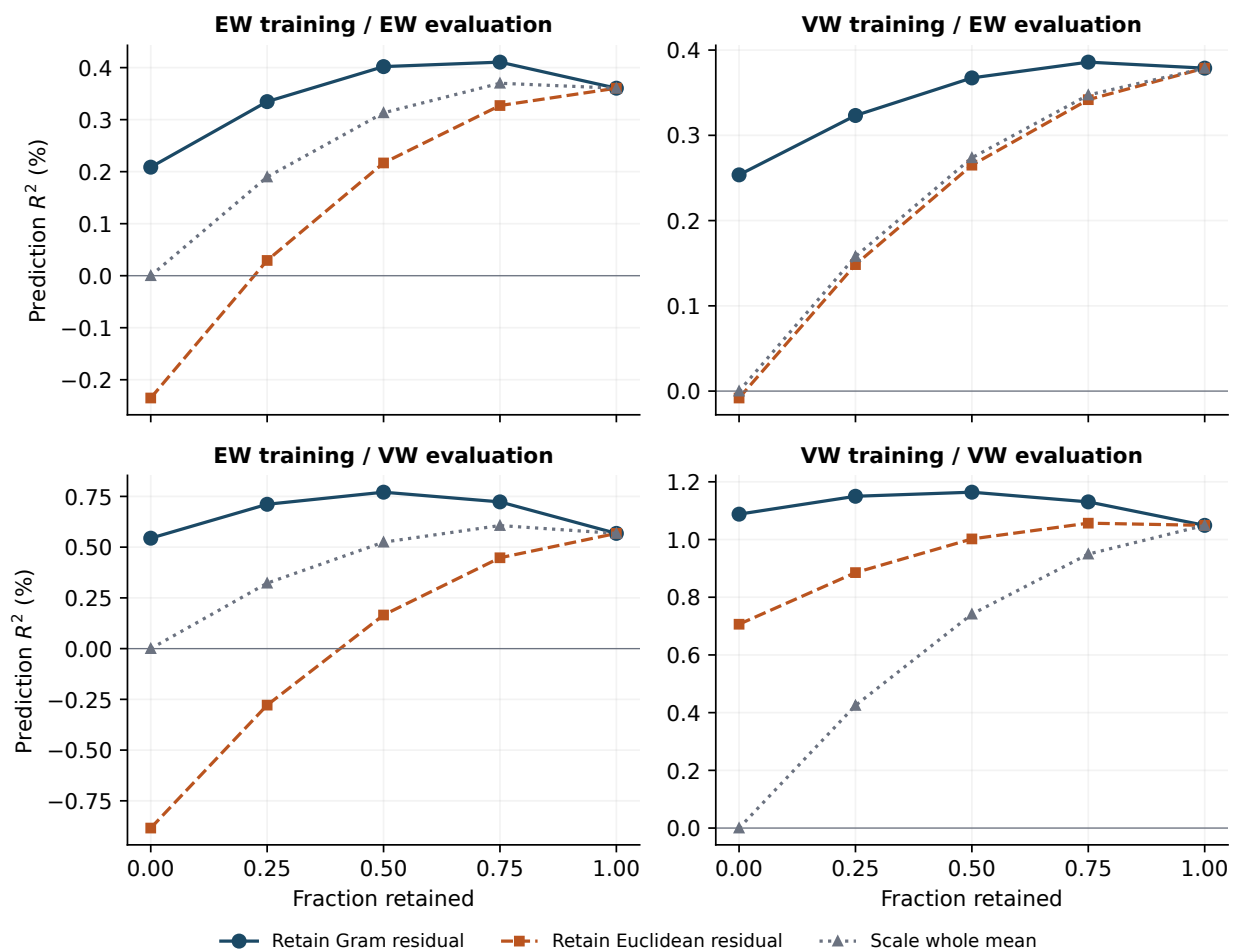


Figure F.1: Which part of the forecast should be scaled?

Each panel holds fixed the three-factor expanding model and evaluation weights. Curves report prediction- R^2 levels in percent over 105 target months, April 2017–December 2025. Gram and Euclidean curves retain the projected mean plus the displayed fraction of its residual. Gram uses the preceding-60-month formation matrix; Euclidean uses the identity in the original rank coordinates. Whole-mean scaling multiplies the entire forecast by that fraction. At one all forecasts coincide; at zero only whole-mean scaling produces the zero forecast. EW and VW denote equal-stock and formation-capitalization weights, with training and evaluation identified in the panel labels. All five fixed fractions are shown, with lines connecting the evaluated rules; different vertical scales are used across panels. The comparison asks whether preserving one component helps beyond reducing the forecast's overall amplitude.

F.2 Past-block scalar estimation

Let $p_{o-1}, r_{o-1} \in \mathbb{R}^L$ be the components actually issued at the preceding origin, and let $\mathcal{H}_{o-1}$ contain the twelve months whose outcomes have now arrived. A scalar minimizing their accumulated loss, constrained to $[0, 1]$, is

$$\hat{\eta}_o = \text{clip}_{[0,1]} \frac{\sum_{t+1 \in \mathcal{H}_{o-1}} r'_{o-1}(X_t - W_t p_{o-1})}{\sum_{t+1 \in \mathcal{H}_{o-1}} r'_{o-1} W_t r_{o-1}}. \quad (\text{F.3})$$

The operator $\text{clip}_{[0,1]}$ truncates its scalar argument to the interval $[0, 1]$. To obtain the formula, differentiate the summed quadratic loss: its derivative is $-2\mathbf{A}_o + 2\eta\mathbf{B}_o$, where the scalar $\mathbf{A}_o$ is the numerator and $\mathbf{B}_o$ the denominator above. The unconstrained minimum is $\mathbf{A}_o/\mathbf{B}_o$ when $\mathbf{B}_o > 0$; truncation then gives the constrained minimum because the objective is convex. The denominator is strictly positive in the recorded data. For a numerically zero denominator the implementation chooses one half, using the tolerance recorded in the code. The new issued forecast is $p_o + \hat{\eta}_o r_o$. Replacing the old residual by the old complete mean and setting the old projection to zero yields whole-mean calibration. Both rules therefore estimate one scalar from the same completed outcomes.

No calibration observation is evaluated as a subsequent outcome for that origin. The first comparable forecast block begins in April 2018, leaving 93 common target months. The saved scalar file records the final calibration return date and the first test date, and the code checks their strict ordering. The identity-ruler follow-up uses its own previously issued components under the identical timing. On the common horizon, its residual-calibrated R^2 values are 0.2595% and 0.2789% under EW evaluation, and -0.0749% and 0.7795% under VW evaluation, for EW and VW training respectively. They are below the corresponding Gram values in all four cells.

A leave-one-origin-out check locates the sensitivity of the gains. Under VW evaluation, the residual-calibration advantage over the full forecast ranges from 0.1391 to 0.2713 percentage points for EW training and from 0.0328 to 0.1671 for VW training. Both remain positive against whole-mean calibration as well. Under EW evaluation, the corresponding advantages over whole-mean calibration range from -0.0209 to 0.0657 and from -0.0118 to 0.0888 points. These are deletion sensitivities, not confidence intervals. The complete eight-row accounting is in `calibration_stability.csv`.

The information sequence makes the adaptive rule feasible within the stored vintage. Its research selection remains exploratory: the calibration and identity-matrix controls were added after inspecting the fixed-fraction curve. The comparison consequently shows how these rules performed on a shared historical record. Assessment of a persistent advantage

requires further evidence on a new period or population and uncertainty analysis for the full procedure.

F.3 Retention, ordinary shrinkage and the ruler

Figure F.1 reports all five fixed fractions 0, 1/4, 1/2, 3/4, 1, alongside the same fractions applied to the entire mean. Under EW training and EW evaluation, prediction R^2 rises from 0.2087% for the projection alone to 0.4020% with half the residual; the complete forecast achieves 0.3605%. Under VW training and VW evaluation, the corresponding values are 1.0879%, 1.1641% and 1.0491%. An interior grid point exceeds both endpoints in all four training/evaluation combinations, but one-half retention does so in only three: under VW training and EW evaluation, three-quarter retention performs better than full retention whereas one-half does not. The result makes the calibration question concrete: using a residual at an intermediate amplitude can outperform both deletion and full retention in this history. Which fraction to issue beforehand must still be determined from an earlier forecasting record.

Two controls clarify what this comparison adds. First, scaling the whole mean by one half yields less predictive accuracy than retaining half the Gram residual in all four combinations. The difference is which component is preserved, rather than only the size of the forecast. Second, repeat the projection using the identity matrix in the original rank coordinates. Its partial-retention forecasts are worse than the corresponding Gram forecasts at each of the four fractions below one in every combination. At one, all constructions coincide with the complete mean. Thus the empirical distinction appears when a user acts on the decomposition, even though decomposing the complete fit alone leaves its predictions unchanged.

For a forecast user, these comparisons concern how much weight to place on two estimated parts of the same expected-return signal. The Gram projection preserves the part closest to the original stock-level signal among coefficients in the loading space. This is a useful measurement property, but its predictive relevance must be assessed using subsequent returns. The main-paper residual-benefit identity shows why neither a large residual nor a small one guarantees a gain. For example, retaining the full residual of the EW-trained model adds 0.1518 percentage points under EW evaluation, whereas full retention for the VW-trained model subtracts 0.0388 points under VW evaluation. The latter sign reverses when some annual blocks are omitted. The signs identify whether the residual improves the stock-return forecast at full strength. Their variation across tasks and blocks explains why the historical calibration and stronger controls are needed before choosing how much of that component

to retain.

G Matched Forecast-Use Challenge

The main calibration comparison asks whether adapting a model’s residual contributes more than generic shrinkage or combination. This appendix makes those alternatives explicit. Every rule is scored on the same subsequent returns, so a difference in loss can be traced to its issued forecast rather than to a different horizon or evaluation population.

G.1 Information and the calibration objective

The stronger benchmark exercise evaluates all rules on 93 target months, April 2018–December 2025. Each origin uses the preceding twelve outcomes of the forecasts actually issued at the previous origin. The code label 201802 refers to the final formation month in the fitted model: its last training and calibration return is March 2018, and the first test return is April 2018. The last block contains nine months. No future outcome chooses a scale, loading space, reference mean or training window within a reported rule. These dates establish the rule’s information sequence. As described in Appendix F, the choice to study the rules followed inspection of the same history, so the performance comparison remains exploratory.

Let $\mathcal{C}_o$ be the twelve calibration formation months at annual origin o . Let $p_o^-, r_o^- \in \mathbb{R}^L$ be the projected and residual coefficient vectors issued at the preceding origin, where $L = 153$. The minus sign denotes the preceding issued forecast, not a coefficient refitted on $\mathcal{C}_o$. For each formation month t , the evaluation moments are $W_t \in \mathbb{R}^{L \times L}$, $X_t \in \mathbb{R}^L$ and scalar $Y_{2,t}$. The matrix W_t is dimensionless, X_t has monthly decimal-return units, and $Y_{2,t}$ has squared monthly-return units. Define the scalar calibration loss

$$\mathcal{L}_o(b) = \sum_{t \in \mathcal{C}_o} (Y_{2,t} - 2X_t' b + b' W_t b), \quad b \in \mathbb{R}^L. \quad (\text{G.1})$$

The residual rule minimizes $\mathcal{L}_o(p_o^- + \eta r_o^-)$ over the dimensionless scalar $\eta \in [0, 1]$. Its solution is Equation (F.3), with the numerator equal to the preceding residual’s alignment with errors of the preceding projection and the denominator equal to its squared signal size. The interval constraint limits the action to retaining some fraction of the issued residual. The next block receives $p_o + \hat{\eta}_o r_o$, where $p_o, r_o \in \mathbb{R}^L$ are the new origin’s components. Thus past outcomes select one amplitude, and current characteristics translate the adjusted coefficient into the next block’s stock forecasts.

The whole-mean rule instead minimizes $\mathcal{L}_o(\gamma h_o^-)$ over $\gamma \in [0, 1]$, where $h_o^- = p_o^- + r_o^-$ is the previous complete mean. Its scalar has the same form with zero base and direction h_o^- , and the next block receives $\hat{\gamma}_o h_o$. This control holds calibration information and the number of fitted parameters fixed while asking whether reducing the entire forecast is as useful as reducing only its residual component.

G.2 Two-component and characteristic-PC controls

Let $V_o = [p_o^- \ r_o^-] \in \mathbb{R}^{L \times 2}$, $A_o = V_o'(\sum_{t \in \mathcal{C}_o} W_t)V_o \in \mathbb{R}^{2 \times 2}$ and $c_o = V_o' \sum_{t \in \mathcal{C}_o} X_t \in \mathbb{R}^2$. The two-component rule lets both the projected and residual signals receive estimated weights. It minimizes $\theta' A_o \theta - 2c_o' \theta$ over $\theta = (u, v)' \in [0, 1]^2$, then issues $up_o + vr_o$. The implementation checks an admissible interior solution, each edge minimum and all corners. Ties select the solution closest to $(1, 1)$, with a deterministic final ordering. A separate bounded least-squares calculation checks the solution, and the implementation also checks objective values and first-order inequalities. Estimating both scales relaxes the restriction that the projected component retains unit weight, but adds estimation noise to a short calibration history.

For the characteristic-PC control, let $U_{K,o} \in \mathbb{R}^{L \times K}$ contain orthonormal eigenvectors associated with the largest K eigenvalues of the positive-definite historical formation Gram $Q_o \in \mathbb{R}^{L \times L}$. Keep the complete mean and measurement population fixed, but replace the estimated factor space by these leading characteristic directions. Project the same current complete coefficient $h_o \in \mathbb{R}^L$ as

$$p_o^{\text{PC}} = U_{K,o}(U_{K,o}' Q_o U_{K,o})^{-1} U_{K,o}' Q_o h_o, \quad r_o^{\text{PC}} = h_o - p_o^{\text{PC}}. \quad (\text{G.2})$$

Use the corresponding previously issued PC components to fit the scalar. The characteristic coordinates and measurement weights are held fixed. The comparison tests whether the fitted factor space supplies a more useful forecast decomposition than a simple space built from characteristic variation. These are characteristic PCs in a fixed encoding, distinct from return PCA and from the native pricing-error estimators discussed in Appendix I.

G.3 Model combination, selection and reporting

Let $h_{E,o}^-, h_{V,o}^- \in \mathbb{R}^L$ denote the previously issued EW and VW coefficients. The convex-combination rule provides an alternative to adjusting a single model: it adapts the weight assigned to two complete forecasts. It minimizes $\mathcal{L}_o(h_{V,o}^- + \omega(h_{E,o}^- - h_{V,o}^-))$ for $\omega \in [0, 1]$, then combines the current coefficients with that weight. Equal averaging fixes $\omega = 1/2$. Past-loss selection chooses the previous lower-loss candidate, with an equal average at a numerical tie.

The common-target selector uses current, historically estimated ridge means and loading spaces, as defined in the main text. Because these combination and selection rules use both candidates, repeated entries in different training columns denote the same issued forecast, not a second estimation or evaluation.

Table G.1 reports all 24 factor-count, window and weighting cells. The replication also reports all thirteen nonduplicated rule labels plus the separate EW/VW component labels, monthly losses, annual and group contributions, fitted scales and timing records. Deleting one evaluation block recalculates its score; it does not refit models or provide a confidence interval. The wider grid asks whether a favorable comparison survives changes to factor count, estimation window and weighting. Its cells use the same underlying history, so their counts describe sensitivity rather than independent replications.

Table G.1: Residual-calibration gains across factor counts and training windows

Window / K	Train	Eval.	Full	Whole	PC	Average
<i>Calibration gain over each baseline (R^2 percentage points)</i>						
Expanding / 1	EW	EW	0.0514	0.0171	-0.0375	-0.0614
	VW	EW	-0.1012	-0.0479	-0.1178	-0.1383
	EW	VW	-0.1120	0.2425	-0.3533	-0.4429
	VW	VW	0.1510	1.2581	0.0713	0.1513
Expanding / 3	EW	EW	0.0387	0.0367	-0.0165	-0.0324
	VW	EW	-0.0001	0.0559	-0.0103	-0.0621
	EW	VW	0.2194	0.4033	-0.0210	-0.1471
	VW	VW	0.1272	1.2127	0.0853	0.2086
Expanding / 5	EW	EW	0.0279	0.0340	-0.0376	-0.0030
	VW	EW	0.0039	0.0339	-0.0238	-0.0922
	EW	VW	0.2428	0.4215	0.0088	-0.0383
	VW	VW	0.1593	1.1313	0.0907	0.1915
Rolling 240 / 1	EW	EW	0.0010	0.0237	-0.0425	0.0741
	VW	EW	0.0082	-0.0278	-0.0413	-0.2512
	EW	VW	-0.4134	0.4197	-0.5973	-0.4810
	VW	VW	0.1120	1.0006	-0.0857	-0.1554
Rolling 240 / 3	EW	EW	0.0114	0.0609	0.0083	0.0914
	VW	EW	0.0122	-0.0390	-0.0258	-0.1998
	EW	VW	0.1083	1.0668	-0.0022	0.0383
	VW	VW	0.0824	1.0664	-0.1295	-0.0739
Rolling 240 / 5	EW	EW	0.0071	0.0777	-0.0095	0.1017
	VW	EW	0.0096	-0.0302	-0.0171	-0.2085
	EW	VW	0.1574	1.0991	0.0013	0.1340
	VW	VW	0.0791	0.9937	-0.0765	-0.0996

Each entry subtracts the named baseline's prediction R^2 from that of factor-space residual calibration, on the same 93 target months, April 2018–December 2025. Positive values favor residual calibration; negative values favor the baseline. Full retains the complete forecast; Whole calibrates its entire amplitude; PC calibrates the residual from the leading K characteristic-Gram eigenvectors; Average equally combines the complete EW/VW forecasts. Train and Eval. identify estimation and loss-measurement weights: EW is equal stock, VW is formation capitalization. Rolling windows contain 240 months. Every adaptive comparison has eight annual calibration decisions using previously issued forecasts. The grid shows sensitivity on a shared sample, including the reversal between expanding and rolling windows under VW training and VW evaluation.

H Additional Geometry and Estimation Controls

H.1 A change in coordinates or a change in estimator?

The residual-retention exercise acts on the decomposition after estimation. A second practical choice acts earlier: a researcher may standardize characteristics before fitting a regularized mean model. Section 2.4 distinguishes a coordinate change from a change in the penalty. Table H.1 measures that distinction using the same annual forecasting task. It restores the economic question behind representation checks: can an apparent improvement be attributed to additional information, or does it come from a different rule for estimating the same signal?

Table H.1: Recoding changes forecasts through the penalty

Recoding	Prediction R^2 (%)		Forecast RMS change (bp)	
	EW eval.	VW eval.	EW eval.	VW eval.
Original ranks	0.3972	0.8550	0.00	0.00
Common rescaling	0.3972	0.8550	0.00	0.00
Orthogonal rotation	0.3972	0.8550	0.00	0.00
RMS standardization	0.4291	0.8054	23.18	4.21
Gram whitening	0.3956	0.5169	52.42	39.13

The first two columns are prediction- R^2 levels in percent. The last two measure the stock-weighted RMS difference from the original-rank ridge forecast in monthly basis points (bp), rather than an R^2 gain. EW evaluation weights stocks equally; VW evaluation uses formation capitalization. Every recoding is nonsingular and estimated from training observations, preserving the available linear signals. Identity penalties in each encoding are retuned on the same training-only grid. RMS standardization divides columns by their training RMS; whitening makes the training Gram the identity. Inner-validation transformations use only inner-fit observations. Transporting the original penalty reproduces its forecast in every encoding, with maximum monthly signal discrepancy below 4×10^{-12} bp. All rows are evaluated on the same 105 target months and clipped returns, showing how the penalty can change forecasts even when attainable signals are unchanged.

The controls are original ranks, common multiplication by ten and a fixed orthogonal rotation. The substantive recodings divide each characteristic by its training RMS or whiten the training Gram. Every map is nonsingular and uses available training data only. For each map, identity-penalty ridge is tuned on the same five-point grid within the training sample. Transformations used for validation are estimated on the earlier inner fit block, and final transformations are re-estimated on the full outer training window. Thus no future Gram or return enters the recoding.

With the original penalty transported, all recodings reproduce the baseline forecast nu-

merically. With a new identity penalty, RMS standardization raises EW prediction R^2 from 0.3972% to 0.4291%, but lowers VW R^2 from 0.8550% to 0.8054%. Whitening gives 0.3956% and 0.5169%, respectively. Common rescaling and rotation leave the trace-scaled ridge solution unchanged. The RMS forecast changes under standardization are 23.2 bp in the EW task and 4.2 bp in the VW task; these are changes in issued signals, not changes in realized portfolio payoffs.

These results explain the economic role of a representation control. Standardization can redistribute regularization across directions without changing the information or the attainable linear forecasts. Whitening in particular makes identity ridge equivalent to proportional shrinkage of the pooled coefficient; it does not add information about future returns. The useful comparison is therefore between estimation rules for the same information set. Equation (16) locates a related distinction inside the factor estimator: characteristic moments can change how the factor history enters the complete fitted mean.

H.2 Updating and ordinary estimation controls

Table H.2 summarizes the annual updating grid. A fixed-space estimator holds the first-origin loading space fixed and re-estimates its intercept and factor history using the current training window. It exceeds full updating in 16 of the 24 factor-count, window, training-weight and evaluation-weight cells, with a median improvement of 0.0057 percentage points in prediction R^2 . Against pooled ridge and monthly ridge it wins 12 of 24 in each comparison. Full updating itself beats pooled ridge in only 10 cells. The apparent case for fixed-space updating weakens once ordinary mean estimation enters the comparison.

An equal average of the full factor forecast and pooled ridge beats the full factor forecast in 20 cells, the ridge forecast in 22, and both components in 18. These findings make forecast combination a useful control for estimation uncertainty: combining two estimates can improve prediction even when neither construction dominates by itself. This average combines a factor forecast with a direct mean estimate. It differs from the equal-stock/value-weighted factor-forecast average in the main results, so the component definitions must be retained when comparing numbers.

Training windows and date conventions matter because they determine how old and recent characteristic-return relations enter the mean. The accompanying tables report all cells, including reversals, and the stock-count-weighted factor-mean sensitivity. The practical finding is that an updating rule should be assessed jointly with ordinary mean estimation and forecast combination. The present comparison does not identify a generally superior updating rule.

Table H.2: Updating rules against direct mean estimation

Comparison	Positive cells	Median gain (pp)
Fixed space versus full update	16/24	0.0057
Fixed space versus pooled ridge	12/24	0.0035
Fixed space versus monthly ridge	12/24	0.0006
Full update versus pooled ridge	10/24	-0.0208
Factor/ridge average versus factor	20/24	0.0559
Factor/ridge average versus ridge	22/24	0.0396

Gains are differences in prediction R^2 , in percentage points (pp), favoring the first named rule when positive. The 24 specifications combine factor counts 1, 3 and 5; expanding or rolling 240-month training; EW or VW training; and EW or VW evaluation. EW weights stocks equally and VW uses formation capitalization. All forecasts use the same 105 target months, April 2017–December 2025, equal-date factor means and clipped excess returns. Fixed space retains the first-origin loading space while re-estimating the intercept and factor history. Full update re-estimates the space as well. The factor/ridge average equally combines a complete factor forecast and pooled ridge trained under the same stock weights; this differs from averaging the two factor forecasts in Table 3. Counts summarize the related specifications on this common sample.

H.3 Why a moving ruler is not sufficient

A controlled simulation clarifies the mechanism without assigning it to the stock data. It holds the loading coefficients known and crosses two changes: rotation of characteristic second moments and state dependence of the factor mean. A synthetic state is observed before prediction. When geometry rotates but the conditional factor mean is constant, the population geometry correction is zero. Estimating separate state means then adds sampling noise. When the factor mean changes across states, conditioning can help even if geometry is constant. Table D.1 reports all four combinations and their complete prediction errors.

These examples separate a descriptive state change from a useful forecasting channel. A rotating characteristic subspace can alter which coefficient directions generate large stock-level exposures. For that change to support a forecast improvement, it must interact with a relevant conditional mean and be estimated well enough to offset additional noise. The empirical results above test particular available rules rather than infer their value from a rotation angle. That is the sense in which the paper asks when geometry matters.

I Interpretation, Comparators and Inferential Scope

I.1 Recent alpha methods answer a distinct question

Zhang (2024) and Choi and Yuan (2025) are essential comparators for pricing-error definitions and inference. The fixed-window projection in this paper measures a specified mean against a specified loading space. It does not estimate all stock-level pricing errors outside the characteristic span. Nor does a mean of monthly characteristic regressions reproduce a complete low-rank, debiased inside/outside-alpha estimator. The contribution here is a controlled empirical evaluation: identify what changes when a reported residual falls, then test the forecasting decisions suggested by that diagnostic.

The benchmark set does not include a verified native implementation of either recent alpha estimator. Accordingly, the results do not rank this procedure against those methods. A direct comparison would match information dates, encoding, samples and target objects, reproduce each native procedure within its stated domain, and then evaluate the complete forecasts on a common task. In particular, regularizing an unavailable monthly inverse changes the estimator; it cannot silently stand in for the original method.

I.2 What the uncertainty statements do and do not establish

The evidence concerns historical forecast decisions and point comparisons of their subsequent losses. Nine annual choices, overlapping expanding samples and common target months do not justify treating the choices as independent Bernoulli trials. Likewise, specification counts cannot be used as independent replications. The leave-one-origin-out ranges show whether a single annual decision drives the reported comparison; they are sensitivity summaries rather than confidence intervals for incremental value.

Complete-procedure inference would have to account for loading-space estimation, factor means, reference-mean tuning, the formation Gram, and the discontinuous choice between candidate models. A monthly loss-series standard error conditional on saved fits does not automatically include those steps. A resampling design must repeat the relevant sequential estimation and tuning under assumptions that justify its approximation. Coverage must be assessed for near ties, weak factors, serial dependence, changing characteristic moments and an unbalanced stock population, in addition to ordinary positive-signal cases.

Residual calibration adds estimation of a historical scalar and truncation to an interval. Inference would also need to distinguish a previously fixed rule from a rule chosen after inspecting the retention curves. The full benchmark grid addresses a different, immediately observable issue: favorable comparisons can reverse when the training window

changes. Reporting these reversals makes the scope of the result visible, without removing research-selection uncertainty.

There is a separate boundary problem for testing zero residual norms. The estimated norm is nonnegative; under a zero population norm it can remain positive because of estimation noise. An ordinary percentile interval need not include the boundary even when the model is refitted in each resample. The historical comparisons are therefore not interpreted as tests rejecting zero population alpha. Algebraic invariance and numerical reproducibility establish that the stated fitted objects are being compared; a valid pricing-error test additionally requires a justified sampling approximation.

I.3 Data validity and an investment extension

The moments retain the stored characteristic and return vintage. Formation ranks and Gram matrices have been rebuilt with the information order described above, but source checks do not fully resolve extreme return records, vendor-vintage differences and delisting treatment. A later audit checked 442,549 comparable next-month records among 442,794 eligible rows in the 93-month horizon: stored lead returns matched the next recorded return at the stated tolerance in all comparable cases. This is an internal timing check, not independent payoff validation. A selected comparison of 52 historical CRSP tail records with current CIZ data found 37 agreements within 10^{-6} and 15 differences. Vintage and return-convention differences remain unresolved; no stored returns were replaced. The audit retains aggregate provenance records and does not redistribute queried stock-level rows. Evaluating on observed returns also does not solve outcome attrition. The reported prediction target is therefore explicitly the commonly clipped excess return in the stored observed-return population.

For investment evaluation, let $w_t \in \mathbb{R}^{N_t}$ denote an implementable vector of portfolio weights chosen at formation, and let $r_{t+1}^{\text{actual}} \in \mathbb{R}^{N_t}$ contain the actual subsequent excess returns including the declared delisting treatment. The gross portfolio payoff is $w_t' r_{t+1}^{\text{actual}}$. Its conditional variance depends on the stock-return covariance matrix $\Sigma_t \in \mathbb{R}^{N_t \times N_t}$ through $w_t' \Sigma_t w_t$. Neither quantity follows from a characteristic Gram norm. A follow-up must specify leverage, financing, shorting, turnover, costs and any beta-neutrality constraint before computing net performance. That experiment would assess a different economic claim from the prediction losses in this paper.

J Replication Boundary and Exhibit Selection

The replication workflow separates reconstruction from aggregate moments from validation of the licensed source panel. Its inputs are common-population and group moments, formation Grams and factor checkpoints. Replay reconstructs losses, projections and calibration; factor refitting additionally repeats initialization, optimization and convergence checks from the moments. The group-construction audit reconciles the aggregate outputs with the licensed panel, and hashed outputs fix the version used in the reported comparisons. These checks serve different purposes: reproducing an exhibit from moments does not independently validate the underlying stock returns. The manuscript’s source package contains the typeset results; it does not redistribute licensed stock records or the separate computational materials.

Exhibits share the declared population, outcome and timing. Results using premium-only forecasts, different characteristic libraries or a separate balanced panel are not combined with them. The reported benchmark grid retains unfavorable cells and all declared sensitivities, so readers can assess the contribution on the same populations, return targets and information dates.

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